UNIVERSITY OF CALIFORNIA
RIVERSIDE

Geometric Structures Arising From SU(n+1,1)-Higgs Bundles

A Dissertation submitted in partial satisfaction
of the requirements for the degree of

Doctor of Philosophy

in

Mathematics

by

Zachary Virgilio

September 2025

Dissertation Committee:

Dr. Brian Collier, Chairperson
Dr. Peter Samuelson
Dr. Stefano Vidussi

The Dissertation of Zachary Virgilio is approved:

Committee Chairperson

University of California, Riverside

Acknowledgements

I would like to thank my advisor, Professor Brian Collier, for his patience, energy, and passion for this subject that all contributed to the completion of my thesis. I would like to thank my family for their encouragement as well. Especially I would like to thank my wonderful wife Jialin for her continual support.

ABSTRACT OF THE DISSERTATION

Geometric Structures Arising From SU(n+1,1)-Higgs Bundles

by

Zachary Virgilio

Doctor of Philosophy, Graduate Program in Mathematics
University of California, Riverside, September 2025
Dr. Brian Collier, Chairperson

Higgs Bundles are known as a useful tool to study the topology of moduli spaces. Only recently was it known that they could provide information about specific representations. This thesis will focus on a class of $\mathsf{SU}(n+1,1)$-Higgs bundles which can provide specific information about their associated representations, as well as construct a family of complex hyperbolic structures on associated manifolds.

We prove that these Higgs bundles correspond to Anosov representations and in the case of $\mathsf{SU}(2,1)$, they parametrize a family of convex cocompact representations that achieve every Toledo invariant and thus lie in every component of the character variety $\mathcal{M}^B(X, \mathsf{SU}(2,1))$, recovering a historical result with new methods. In addition, we prove analagous results for a class of $\mathsf{SO}_0(2, n+2)$-Higgs bundles, namely, that they correspond to Anosov representations and can be used to parametrize a class of geometric structures on specific manifolds.

Contents

Chapter 1

Introduction

1.1 Overview

In 1987, Nigel Hitchin introduced Higgs bundles as a solution to the Hitchin equations, a reduction of the Yang-Mills self-duality equations to three dimensions in two key papers [Hit87a] and [Hit87b]. The theory of Higgs bundles was developed further by Hitchin, along with Simpson [Sim88], Corelette [Cor88] and Donaldson [Don87] and other contributors. This body of work is now called the non-abelian Hodge correspondence and it gives a homeomorphism between the moduli space of Higgs bundles and the character variety. In particular, for a closed Riemann surface X, that is, a surface with a complex structure, of genus two or more, there is a homeomorphism between the moduli space of G-Higgs bundles, $\mathcal{M}_{\text{Higgs}}(\mathsf{G})$, and the moduli space of reductive representations $\mathcal{X}(\pi_1(X), \mathsf{G})$.

The study of these moduli spaces has gone in several, related, directions. One of the directions that is particularly relevant to this thesis is understanding dynamical properties of a representation from the associated Higgs bundles. For a surface σ of genus $g \geq 2$, a representation into $\mathsf{PSL}(2,\mathbb{R})$ is Fuchsian if it's image is a discrete subgroup. There is a component of $\mathcal{X}(\pi_1(\Sigma), \mathsf{PSL}(2,\mathbb{R}))$ consisting solely of Fuchsian representations, which is identified with Teichmüller space via the isomorphism $\Sigma \cong \mathbb{H}/\pi_1(\Sigma)$. Hitchin [Hit92] identified a component of the character variety that is contractible and contains a copy of Teichmüller space of a Lie group H which is the adjoint form of a simple, split real Lie group. The study of similar components, especially for groups of rank higher than 1, has been termed Higher Teichmüller theory. For a real rank 1 Lie group, convex cocompact representations are a generalization of Fuchsian representations and encompass examples such as quasi-Fuchsian representations of surface groups into $\mathsf{PSL}(2,\mathbb{C})$. However, a result of Kleiner and Leeb [KL05] showed that in higher rank, convex cocompactness is a rigid condition and only yields products of uniform lattices with rank one convex cocompact groups. The appropriate generation to higher ranks was developed by Labourie [Lab06] and called an Anosov representation, based on the idea of Anosov flows. These ideas were developed by a number of authors, including Guichard and Weinhard [GW12], Kapovich, Leeb and Porti [KLP16a, KLP17, KLP16b] and Zhu [Zhu20]. Understanding the properties of Higgs bundles that correspond to these properties has been an active area of research.

Another direction is in the study of geometric structures on manifolds. These were introduced by Cartan and Ehresmann in the 1920s, and popularized by Thurston in his study of the geometrization conjecture. In 2010, Baraglia showed in his thesis [Bar09] that Higgs bundles could be an important tool to construct interesting geometric structures. In particular, the holonomy of a flat bundle corresponds to the holonomy representation of the geometric structure, and a transverse section of the bundle stores the data of the developing map. Baraglia showed how the flat connection can be expressed in terms of the solutions of

Hitchin's equations and how to construct the transverse section from the holomorphic data of the vector bundle. More recently, Alessandrini and Li [ADL24], Collier, Tholozan and Toulisse [CTT19] and have all found further geometric structures which are parametrized by Higgs bundles.

Before Baraglia's thesis, there had been work relating higher Teichmüller theory to geometric structures. In particular, the work of Choi and Goldman [CG93] and Guichard and Weinhard [GW08] related these two ideas. These were later generalized and put in the context of Anosov representations [GW12, KLP17]. Higgs bundles were not involved in this research as early perceptions were that they were strong tools for understanding the topology of the moduli spaces, but that they provided little to no geometric information about a specific point in the representation variety.

A notable breakthrough in this field was the work of Simion Filip on variations of Hodge structure [Fil22]. A variation of Hodge structure can be interpreted as a fixed point of $\mathbb{C}^*$-action on the moduli space of Higgs bundles. In this language, among other results, he showed that with a very specific assumption on the structure of the bundle and Higgs field, this class of $\mathsf{SO}(2,3)$-Higgs bundles are Anosov. A key introduction he made was a specific function that relates to both the Lie theoretic data, and the dynamical data of the representation. He was also able to use the function in an important step in constructing a developing map that assigned a geometric structure to specific bundles over the base Riemann surface X.

This thesis generalizes the methods Filip used to more classes of Higgs bundles. In particular there is an invariant called the Toledo invariant and for $\mathsf{SU}(2,1)$, it parametrizes the components of $\mathcal{X}(\pi_1(\Sigma), \mathsf{SU}(2,1))$. It will be shown that there is class of $\mathsf{SU}(2,1)$-Higgs bundles whose representations are all convex cocompact and that achieve every possible Toledo invariant, thus, that every component of the character variety contains convex cocompact representations.

1.2 Outline and Statement of Results

The results in this thesis involve Lie theory, geometric structures, and Higgs bundles. With this in mind, Chapter 2 will summarize the relevant tools of Lie theory and geometric structure that will be used in the rest of thesis. This includes the KAK decomposition, real forms and complexification, restricted roots systems, developing maps and holonomy of geometric structures. In Chapter 3 we will present an overview of the nonabelian Hodge correspondence. This is primarily intended to highlight the way Higgs bundles interact with the character variety, and which structures they inherit as a result of stability. The details of the nonabelian Hodge are beyond the scope of this work, so we will take these results for granted. We will spend more time focusing on Higgs bundles. In this chapter, we will also introduce the notion of Anosov representation in a modern context that avoids the Anosov flows of Labourie's [Lab06] original defintion.

In Chapter 4, we will take a detour to prove some technical results that generalize Filip's work. The key function he constructs satisfies a few conditions regarding it's gradient and we will organize these properties into a general class of functions and then prove the main results that we will use in a more general context. Additionally, we provide specific details in the proofs that were lacking and for which a source could not be found in a (brief) search of literature. The key result provde in this chapter is on the exponential growth of the function.

Chapters 5 and 6 are where we prove the main results of this thesis. Each chapter will begin with the specifics of the Lie groups we will focus on in that chapter, so $\mathsf{SU}(n+1,1)$ and $\mathsf{SO}(2,n+2)$ respectively. We will also introduce the associated geometric structures and specify the conditions on the class of Higgs bundle we consider. In both chapters, we will consider Higgs bundles that decompose into subbundles where there are two specific line bundles such that the Higgs fields maps one line bundle to the other in a non-vanishing manner, and fixes the second. Diagrammatically, it would have the form:

$$\cdots \longrightarrow U \xrightarrow{1} V$$

Where U and V are the line bundles and 1 indicates a nonvanshing map that is part of the data of the Higgs field. Notice that the Higgs field does not send V anywhere, so it is a fixed subbundle.

Remark 1.1. The requirement that the Higgs field not map $\mathcal{V}$ anywhere can be loosened. Junming Zhang showed that in the case of parabolic $\mathsf{SO}_0(2,3)$-Higgs bundles, the Higgs field can map $\mathcal{V}$ back to the bundle and that these bundles still correspond to Anosov representations [Zha24].

Throughout the thesis, we will have a fixed Riemann surface X of genus $g \geq 2$. In Chapter 5, our first main result will be on the existence of bundles over X with complex hyperbolic and boundary complex hyperbolic geometries. We will work with $\mathsf{SU}(n+1,1)$-Higgs bundles parametrized by a triple $(\mathcal{V}_1, \mathcal{V}_2, \beta)$, where $\mathcal{V}_1$ is a rank n holomorphic vector bundle, $\mathcal{V}_2$ is a holomorphic line bundle and β is a holomorphic 1-form valued in $\hom(\mathcal{V}_1, \mathcal{V}_2)$. We construct a Higgs bundle from this data, which is represented schematically as:

$$\mathcal{V}_1 \xrightarrow{\beta} \mathcal{V}_2 \xrightarrow{1} \mathcal{V}_3$$

where 1 is the identity map from $\mathcal{V}_2 \to \mathcal{V}^2 K^{-1} \otimes K = \mathcal{V}_2$. The details are expaned upon in Chapter 5, Section 5.3.

From the structure of these Higgs bundle, we will construct a family of functions satisfying the conditions of Chapter 4. Using this, we will show the existence of an S^{2n-1}-bundle over $\widetilde{X}$, $\widetilde{\mathcal{S}} \subset \widetilde{X} \times \partial\mathbb{CH}^{n+1}$ and a B^{2n}-bundle also over $\widetilde{X}$, $\widetilde{\mathcal{D}} \subset \widetilde{X} \times \mathbb{CH}^{n+1}$. We will identify these bundles with subbundles of the Higgs bundles that are our focus and we will also show that they descend in an equivariant manner to bundles $\mathcal{S}$ and $\mathcal{D}$ over X. We then define developing maps that will equip these bundles with geometric structures:

$$\mathrm{Dev}^S : \widetilde{\mathcal{S}} \longrightarrow \partial\mathbb{CH}^{n+1}$$

$$\mathrm{Dev}^D : \widetilde{\mathcal{D}} \longrightarrow \mathbb{CH}^{n+1}$$

which are both given by projection on the second factor. With the notation $\pi_1(X) = \Gamma$, we have our first pair of results.

Theorem 5.7. *There exists a unique open set $\Omega_\rho \subset \partial\mathbb{CH}^{n+1}$ such that Dev^S is a diffeomorphism between $\widetilde{\mathcal{S}}$ and Ω_ρ. Then Dev^S is a developing map and so induces a $\partial\mathbb{CH}^{n+1}$ structure on $\mathcal{S}$.*

Moreover, the action of Γ on Ω_ρ is properly discontinuous, and Ω_ρ/Γ is diffeomorphic to $\mathcal{S}$ and $\mathcal{S}$ is diffeomorphic to a S^{2n-1}-bundle.

Theorem 5.8. *The map* Dev^D *is a diffeomorphism between* $\widetilde{\mathcal{D}}$ *and* $\mathbb{CH}^{n+1}$ *and so* Dev^D *is a developing map and induces a* $\mathbb{CH}^{n+1}$ *structure on* $\mathcal{D}$.

Moreover, the action of Γ *on* $\mathbb{CH}^{n+1}$ *is properly discontinuous and* $\mathbb{CH}^{n+1}/\Gamma$ *is diffeomorphic to* $\mathcal{D}$*, which is diffeomorphic to a complex disk bundle* B^{n+1} *over* X*.*

The statements of these theorems are very similar, and that is reflected in their proofs. The techniques are similar and in both cases, a key step in showing injectivity (and in the case of Theorem 5.8 surjectivity), relies on properties of the special functions we consider in Chapter 4. The idea for these proofs was heavily inspired by Filip in [Fil22], in particular, his observation that uniqueness of minima of the function relates to the developing map. The rest of the proof consists showing that the developing map is a local diffeomorphism, which is proved in local coordinates.

We then prove a technical result that will be used in the proof of the Anosov property for both $\mathsf{SU}(n+1,1)$- and $\mathsf{SO}(2,n+2)$-Higgs bundles. We pullback our Higgs bundle to a bundle over $\widetilde{X}$, and then it bounds the growth of the vector with respect to the metric at γx_0 for $\gamma \in \pi_1(X)$ by an exponential function that depends on the choice of root. Armed with this inequality, we able to prove the Anosov property for $\mathsf{SU}(n+1,1)$-Higgs bundles, combining the aforementioned exponential bound on the length of a vector with the exponential growth condition that was a highlight of Chapter 4.

Theorem 5.10. *For a polystable* $\mathsf{SU}(n+1,1)$*-Higgs bundle that decomposes as*

$$\mathcal{V}_1 \xrightarrow{\beta_1} \mathcal{V}_2 \xrightarrow{1} \mathcal{V}_3$$

the associated representation $\rho : \pi_1(X) \to \mathsf{SU}(n+1,1)$ is Anosov. Moreover, Ω_ρ, from Theorem 5.7, is a domain of discontinuity and the complement of the open set Ω_ρ in $\partial\mathbb{CH}^{n+1}$ coincides with the limit set Λ_ρ.

And since $\mathsf{SU}(n+1,1)$ is rank 1, there is an immediate corollary.

Corollary 5.11. *With the same set up as in 5.10, the representation ρ is convex cocompact.*

We then show that there is a parametrization of the class of bundles from Theorem 5.10.

Theorem 5.12. *For $\mathsf{SU}(2,1)$ and a choice of Riemann surface structure X on a surface of genus $g \geq 2$, there is a parametrization of a class of convex cocompact representations from Theorem 5.11 given by a line bundle L and a section $\beta \in H^0(X, \hom(L^2K, L^{-1}) \otimes K)$.*

This result combines with established facts about the character variety of $\mathsf{SU}(2,1)$ to recover an important result of Goldman, Kapovich and Leeb [GKL01] using these new techniques:

Corollary 5.13. *For each possible value of the Toledo invariant on the character variety of $\mathsf{SU}(2,1)$, there is a convex cocompact representation with that value. In particular, there is a convex cocompact representation in every component of the character variety.*

In Chapter 6, we will follow a similar structure and prove analagous results. We will begin with the same setup: A Riemann surface X of genus $g \geq 2$ and a $\mathsf{SO}_0(2,n+2)$-Higgs bundle over X with an associated representation ρ. The class of polystable $\mathsf{SO}_0(2,n+2)$-Higgs bundles that are relevant to use are parametrized by a triple $(\mathcal{W}_3, \mathcal{W}_4, \alpha)$ where $\mathcal{W}_3$ is a degree 0, rank n holomorphic vector bundle, $\mathcal{W}_4$ is a holomorphic line bundle and α is a section in $H^0(X, \hom(\mathcal{W}_3, \mathcal{W}_4) \otimes K)$. We then define the line bundle $\mathcal{W}_2 = \overline{\mathcal{W}_4}$, and line bundles $\mathcal{W}_5 = \mathcal{W}_4 K^{-1}$ with $\mathcal{W}_1 = \overline{\mathcal{W}_5}$. In a diagram, we can represent the Higgs bundles as:

$$\mathcal{W}_1 \xrightarrow{1} \mathcal{W}_2 \xrightarrow{\alpha^T} \mathcal{W}_3 \xrightarrow{\alpha} \mathcal{W}_4 \xrightarrow{1} \mathcal{W}_5$$

where 1 is the identity map from $\mathcal{V}_4 \to \mathcal{V}_4 K^{-1} \otimes K$. The full details are given in Chapter 6, Section 6.2.

We can again construct two bundles over X which will be given geometric structures relating to isotropic and positive lines in the Higgs bundle.

Theorem 6.4. *There exists a unique open set $\Omega_\rho \subset \mathrm{Ein}^{1,n+1}$ such that Dev^B is a diffeomorphism between $\widetilde{\mathcal{B}}$ and Ω_ρ. Then Dev^B is a developing map and defines a $\mathrm{Ein}^{1,n+1}$ structure on $\mathcal{B}$.*

Moreover, the action of Γ on Ω_ρ is properly discontinuous, and Ω_ρ/Γ is diffeomorphic to $\mathcal{B}$, where in turn, $\mathcal{B}$ is diffeomorphic to a S^{n-1}-bundle.

Theorem 6.5. *The map Dev^P is a diffeomorphism between $\widetilde{\mathcal{P}}$ and $\mathbb{PS}^{1,n+1}$ and so Dev^P is a developing map which gives $\mathcal{P}$ a $\mathbb{PS}^{1,n+1}$ geometric structure.*

Additionally, Γ acts properly discontinuously on $\mathbb{PS}^{1,n+1}$ and $\mathbb{PS}^{1,n+1}$ is diffeomorphic to $\mathcal{P}$.

The proofs for these theorems will follow a very similar structure to Theorems 5.7 and 5.8. We will use the class of functions constructed using the Higgs bundle so that they belong to the family of functions discussed in Chapter 4. These will show the injectivity of the developing maps in both cases, and surjectivity in Theorem 6.5. Then local calculations will prove the maps are diffeomorphisms. There proof of Theorem 5.10 is easily adaptable to the case of $\mathsf{SO}_0(2, n+2)$ and so there is an accompanying result.

Theorem 6.6. *Let E be a polystable $\mathsf{SO}_0(2, n+2)$-Higgs bundle that decomposes as*

$$\mathcal{W}_1 \longrightarrow \mathcal{W}_2 \longrightarrow \mathcal{W}_3 \xrightarrow{\alpha} \mathcal{W}_4 \xrightarrow{1} \mathcal{W}_5$$

Then the associated representation $\rho : \pi_1(X) \to \mathsf{SO}_0(2, n+2)$ is α_2-Anosov and Ω_ρ from Theorem 6.4 is a domain of discontinuity.

Chapter 2

Lie Theory and Geometric Background

2.1 Lie Algebra Basics

Cartan Subalgebras and Rank

If $\mathfrak{g}$ is a finite dimensional complex Lie algebra, not necessarily semisimple, and $\mathfrak{h}$ is a nilpotent Lie subalgebra, then the *generalized weight spaces of* $\mathfrak{g}$ *relative to* $\mathrm{ad}_{\mathfrak{g}}\mathfrak{h}$ are defined as

$$\mathfrak{g}_\alpha = \{X \in \mathfrak{g} \mid (\mathrm{ad}\, H - \alpha(H)1)^n X = 0 \text{ for all } H \in \mathfrak{h} \text{ and some } n \text{ that depends on } X \text{ and } H\}$$

These generalized weight spaces satisfy the following conditions:

1. $\mathfrak{g} = \bigoplus \mathfrak{g}_\alpha$

2. $\mathfrak{h} \subseteq \mathfrak{g}_0$, where $\mathfrak{g}_0$ is the generalized weight space for the zero weight

3. $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$ with the assumption that $\mathfrak{g}_{\alpha+\beta}$ is 0 whenever $\alpha+\beta$ is not a generalized weight.

From 3, it is clear that $\mathfrak{g}_0$ is a subalgebra.

Definition 2.1. For a complex Lie algebra $\mathfrak{g}$ of finite dimension, a nilpotent Lie subalgebra $\mathfrak{h}$ is a *Cartan subalgebra* of $\mathfrak{g}$ if $\mathfrak{h} = \mathfrak{g}_0$.

A Cartan subalgebra will always be abelian.

We recall the generalized weight spaces were denoted as $\mathfrak{g}_\alpha$ for α in $\mathfrak{h}^*$. A *root* of $\mathfrak{g}$ with respect to $\mathfrak{h}$ is an α for which $\mathfrak{g}_\alpha$ is non-zero. The set of roots is denoted as $\Delta(\mathfrak{g}, \mathfrak{h})$, or when there is no ambiguity about the Lie algebra and Cartan subalgebra, simply as Δ. We can then rewrite the generalized weight space decomposition of $\mathfrak{g}$ as

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha$$

which is known as the *root-space decomposition* with respect to $\mathfrak{h}$. Also, an element of $\mathfrak{g}_\alpha$ is called a *root vector* for the root α.

An important result guarantees that any finite dimensional complex Lie algebra has a Cartan subalgebra. Furthermore, the Cartan subalgebra is abelian. In fact, in a complex semisimple Lie algebra $\mathfrak{g}$, a Cartan subalgebra can be characterized as being the maximal abelian subalgebra $\mathfrak{h}$ such that $\mathrm{ad}_{\mathfrak{g}} \mathfrak{h}$ is simultaneously diagonalizable.

For a finite dimensional real semisimple Lie algebra $\mathfrak{g}$, a Cartan subalgebra $\mathfrak{h}$ is a Lie subalgebra satisfying the condition that $\mathfrak{h}^{\mathbb{C}}$ is a Cartan subalgebra of $\mathfrak{g}^{\mathbb{C}}$.

For a complex Lie algebra $\mathfrak{g}$, a Cartan subalgebra is unique up to conjugation by an automorphism of $\mathfrak{g}$. Since all Cartan subalgebras have the same dimension, this gives rise to the *rank* of a Lie algebra $\mathfrak{g}$, where the rank is the dimension of a Cartan subalgebra.

Complexification and Real Forms

Let $\mathfrak{g}$ be a real Lie algebra. As a vector space, the operation of tensoring $\mathfrak{g}$ with $\mathbb{C}$ yields a vector space $\mathfrak{g} \otimes \mathbb{C}$ over the complex numbers. This operation extends to the Lie algebra structure on $\mathfrak{g}$ as well: consider the map

$$(X, z, Y, w) \in \mathfrak{g} \times \mathbb{C} \times \mathfrak{g} \times \mathbb{C} \mapsto [X, Y] \otimes zw \in \mathfrak{g} \otimes \mathbb{C}$$

This map is linear in each factor, so the universal property of tensors allows us to extend it to a linear map on $\mathfrak{g} \otimes \mathbb{C} \otimes \mathfrak{g} \otimes \mathbb{C}$. We then restrict it to a bilinear map on $(\mathfrak{g} \otimes \mathbb{C}) \times (\mathfrak{g} \otimes \mathbb{C})$ which satisfies the conditions of a Lie bracket on $\mathfrak{g} \otimes \mathbb{C}$. We call this new complex Lie algebra the *complexification of* $\mathfrak{g}$ and denote it by $\mathfrak{g}^{\mathbb{C}}$.

It is also possible to restrict the scalars of a complex Lie algebra $\mathfrak{g}$ to $\mathbb{R}$. This operation of shrinking the scalar field is denoted by $\mathfrak{g}^{\mathbb{R}}$.

When $\mathfrak{g}$ is a complex Lie algebra and there is a real Lie algebra $\mathfrak{g}_0$ such that

$$\mathfrak{g}^{\mathbb{R}} = \mathfrak{g}_0 \oplus i\mathfrak{g}_0$$

as vector spaces, we call $\mathfrak{g}_0$ a *real form* of $\mathfrak{g}$. Any real Lie algebra is then a real form of its complexification. This gives rise to the operation of *conjugation with respect to a real form*: the $\mathbb{R}$-linear map $1 \oplus -1$ on $\mathfrak{g}_0 \oplus i\mathfrak{g}_0$.

There are special real forms which are important to the structure of a semisimple Lie algebra. One notion is that of a split real form. If $\mathfrak{g}$ is a complex semisimple Lie algebra, and $\mathfrak{h}$ is a Cartan subalgebra, and Δ the roots of $\mathfrak{g}$ with respect to $\mathfrak{h}$, then define

$$\mathfrak{h}_0 = \{H \in \mathfrak{h} \mid \alpha(H) \in \mathbb{R} \text{ for all } \alpha \in \Delta\}$$

We say that a real form $\mathfrak{g}_0$ of $\mathfrak{g}$ is a *split real form* if $\mathfrak{h}_0$ is contained in $\mathfrak{g}_0$. Every complex semisimple Lie algebra contains a split real form.

Another important class of real forms are compact real forms. As the name suggests, if $\mathfrak{g}$ is a complex semisimple Lie algebra, then a real form $\mathfrak{u}_0$ of $\mathfrak{g}$ that is also a compact Lie algebra is a *compact real form of* $\mathfrak{g}$.

Example 2.2. For some classical Lie algebras, their compact real forms are already familiar.

1. $\mathfrak{su}(n)$ is a compact real form of $\mathfrak{sl}(n, \mathbb{C})$

2. $\mathfrak{so}(n)$ is a compact real form of $\mathfrak{so}(n, \mathbb{C})$

3. $\mathfrak{sp}(n)$ is a compact real form of $\mathfrak{sp}(n, \mathbb{C})$.

An important result is that every complex semisimple Lie algebra possesses a compact real form.

2.1.1 Cartan Decomposition

We begin with a real semisimple Lie algebra $\mathfrak{g}$. To any Lie algebra, it is possible to define a symmetric bilinear form, called the Killing form of $\mathfrak{g}$, from $B : \mathfrak{g} \times \mathfrak{g} \to \mathbb{R}$ by the operation:

$$B(X, Y) := \mathrm{Tr}(\mathrm{ad}\, X\, \mathrm{ad}\, Y).$$

For an element X in $\mathfrak{g}$, the negative conjugate transpose map $X \mapsto -X^*$ is an involution; applying it twice yields the identity and it is a Lie algebra automorphism. This map has a special interaction with the Killing form. It can be used to define a new symmetric, positive definite bilinear form $B'(X, Y) = -B(X, -Y^*)$.

Generalizing this property leads to the following definition:

Definition 2.3. An involution θ of $\mathfrak{g}$ such that the symmetric bilinear form

$$B_\theta(X, Y) := -B(X, \theta Y)$$

is positive definite is called a *Cartan involution* of $\mathfrak{g}$.

Any real semisimple Lie algebra has a Cartan involution and it is unique up to conjugation via $\mathrm{Int}\, \mathfrak{g}$.

Every compact real form $\mathfrak{u}_0$ of a complex semisimple Lie algebra $\mathfrak{g}$ has an associated involution, conjugation with respect to $\mathfrak{u}_0$. This involution is a Cartan involution on $\mathfrak{g}^{\mathbb{R}}$. Since any two Cartan involutions are conjugate via $\mathrm{Int}\, \mathfrak{g}$, it follows that any two compact real forms are also conjugate.

For a real semisimple Lie algebra $\mathfrak{g}_0$ and a Cartan involution θ, there is a corresponding deocmposition into the $+1$ and -1 eigenspaces:

$$\mathfrak{g}_0 = \mathfrak{l}_0 \oplus \mathfrak{p}_0$$

Since θ is an involution, these component spaces follow the bracket conditions:

$$[\mathfrak{l}_0, \mathfrak{l}_0] \subseteq \mathfrak{l}_0, \quad [\mathfrak{l}_0, \mathfrak{p}_0] \subseteq \mathfrak{p}_0, \quad [\mathfrak{p}_0, \mathfrak{p}_0] \subseteq \mathfrak{l}_0$$

An immediate consequence is the orthogonality of the decomposition.

Lemma 2.4. *For a real semisimple Lie algebra $\mathfrak{g}_0$ and Cartan involution θ, the Cartan decomposition is orthognal with respect to the Killing form B and B_θ.*

Proof. For $X \in \mathfrak{l}_0$ and $Y \in \mathfrak{p}_0$, we have $\operatorname{ad} Y(\mathfrak{l}_0) \in \mathfrak{p}_0$ so $\operatorname{ad} X(\operatorname{ad} Y(\mathfrak{l}_0)) \in \mathfrak{p}_0$, so $(\operatorname{ad} X \operatorname{ad} Y)(\mathfrak{l}_0) \in \mathfrak{p}_0$. Similarly, $(\operatorname{ad} X \operatorname{ad} Y)(\mathfrak{p}_0) \in \mathfrak{l}_0$. This means that $\operatorname{ad} X \operatorname{ad} Y$ takes $\mathfrak{l}_0$ to $\mathfrak{p}_0$ and $\mathfrak{p}_0$ to $\mathfrak{l}_0$.

What this means is that any eigenvector of $\operatorname{ad} X \operatorname{ad} Y$ can be written as $u+v$ where $u \in \mathfrak{l}_0$ and $v \in \mathfrak{p}_0$ and neither are zero. So $(\operatorname{ad} X \operatorname{ad} Y)(u+v) = \lambda(u+v)$. Taking this, together with the fact that $\operatorname{ad} X \operatorname{ad} Y$ carries $\mathfrak{l}_0$ to $\mathfrak{p}_0$ and vice-versa, we know that $(\operatorname{ad} X \operatorname{ad} Y)u = \lambda v$ and $(\operatorname{ad} X \operatorname{ad} Y)v = \lambda u$.

So

$$(\operatorname{ad} X \operatorname{ad} Y)(u - v) = \lambda v - \lambda u = -\lambda(u - v)$$

So for any eigenvalue λ, there must be a second eigenvalue $-\lambda$. So the sum of the eigenvalues will be 0.

Thus $\mathrm{Tr}(\mathrm{ad}\, X\, \mathrm{ad}\, Y) = 0$, so $B(X, Y) = 0$.

Since $\theta Y = -Y$, $B_\theta(X, Y) = -B(X, \theta Y) = -B(X, -Y) = B(X, Y) = 0$, so the decomposition into $\mathfrak{k}_0$ and $\mathfrak{p}_0$ is orthogonal with respect to B and B_θ. □

Another fact we see is that since B_θ is positive definite, we know that B must be negative definite on $\mathfrak{k}_0$, since if $X \in \mathfrak{k}_0$, we have:

$$-B(X, \theta X) = -B(X, X) \geq 0$$

so $B(X, X) \leq 0$. Similarly, B must be positive definite on $\mathfrak{p}_0$.

Any decomposition of $\mathfrak{g}_0 = \mathfrak{k}_0 \oplus \mathfrak{p}_0$ which satisfies the conditions:

1.
$$[\mathfrak{k}_0, \mathfrak{k}_0] \subseteq \mathfrak{k}_0, \quad [\mathfrak{k}_0, \mathfrak{p}_0] \subseteq \mathfrak{p}_0, \quad [\mathfrak{p}_0, \mathfrak{p}_0] \subseteq \mathfrak{k}_0$$

2. B is negative definite on $\mathfrak{k}_0$ and positive definite on $\mathfrak{p}_0$.

is called a *Cartan decomposition* of $\mathfrak{g}_0$

There is a correspondence between Cartan involutions and Cartan decompositions. We have seen how any Cartan involution yields a Cartan decomposition. The reverse is straightforwards. For a Cartan decomposition $\mathfrak{g}_0 = \mathfrak{k}_0 \oplus \mathfrak{p}_0$, define the map $\theta : \mathfrak{g}_0 \to \mathfrak{g}_0$ by $\theta(u + v) = u - v$ for $u \in \mathfrak{k}_0$ and $v \in \mathfrak{p}_0$.

For a complex semisimple Lie algebra $\mathfrak{g}$ with compact real form $\mathfrak{u}_0$, we can write the $\mathfrak{g}$ restricted to a real Lie algebra $\mathfrak{g}^{\mathbb{R}}$ as $\mathfrak{g}^{\mathbb{R}} = \mathfrak{u}_0 \oplus i\mathfrak{u}_0$. Then conjugation with respect to this splitting is a Cartan involution. In fact, conjugation with respect to compact real forms are the only Cartan involutions of $\mathfrak{g}^{\mathbb{R}}$. The complexification of the associated Cartan decompositions,

$$(\mathfrak{g}^{\mathbb{R}})^{\mathbb{C}} = \mathfrak{u}_0^{\mathbb{C}} \oplus i\mathfrak{u}_0^{\mathbb{C}} = \mathfrak{u}_0^{\mathbb{C}} \oplus \mathfrak{u}_0^{\mathbb{C}}$$

is called the *complexification of the Cartan decomposition*. This is a key fact when constructing Higgs bundles of complex Lie groups.

The Cartan decomposition of a Lie algebra extends to the level of the Lie groups. For a semisimple Lie group G with Lie algebra $\mathfrak{g}_0$, let θ be a Cartan involution and $\mathfrak{g}_0 = \mathfrak{l}_0 \oplus \mathfrak{p}_0$ be the corresponding Cartan decomposition. Let K be the subgroup of G with Lie algebra $\mathfrak{l}_0$. Then there is an automorphism Θ of G whose differential is θ and $\Theta^2 = 1$. K is the subgroup of G fixed by Θ and G decomposes into

$$G = K \exp \mathfrak{p}_0$$

Furthermore, K contains the center of G, and if the center is finite, then K is a maximal compact subgroup of G. This decomposition generalizes the polar decomposition of matrices.

2.1.2 KAK Decomposition

We now study a second global decomposition of a Lie algebra will be a key component of our main result. It is known as the KAK decomposition and unlike the global Cartan decomposition, it decomposes a Lie group into closed subgroups.

We will let G be a semisimple Lie group and $\mathfrak{g}$ it's Lie algebra. We will fix a Cartan involution θ and a corresponding Cartan decomposition $\mathfrak{g} = \mathfrak{l} \oplus \mathfrak{p}$. Then let K be the subgroup of G with Lie algebra $\mathfrak{l}$. We will use the bilinear form B_θ relative to the Cartan involution when we refer to adjoint and orthogonality.

Now we can compute $(\operatorname{ad} X)^*$ for any $X \in \mathfrak{g}$. We start with $X, Y, Z \in \mathfrak{g}$:

$$B_\theta((\operatorname{ad}\theta X)Y, Z) = -B([\theta X, Y], \theta Z) = -B(-[Y, \theta X], \theta Z) = B(Y, [\theta X, \theta Z])$$

where we used interaction between the Killing form and the Lie bracket. We can also use the property that θ is a Lie algebra automorphism, so $[\theta X, \theta Z] = \theta([X, Z])$. This gives us:

$$B_\theta((\operatorname{ad}\theta X)Y, Z) = B(Y, \theta[X, Z]) = -B_\theta(Y, (\operatorname{ad} X)Z) = B_\theta(-(\operatorname{ad} X)^* Y, Z)$$

So we see that $\operatorname{ad} X^* = -\operatorname{ad}\theta X$.

We now choose a maximal abelian subspace of $\mathfrak{p}$ and denote it by $\mathfrak{a}$. For $H \in \mathfrak{a}$, we see that $\operatorname{ad} H^* = -\operatorname{ad}\theta H = -\operatorname{ad}(-H) = \operatorname{ad} H$. Thus

$$\{\operatorname{ad} H \mid H \in \mathfrak{a}\}$$

is a commuting family of self-adjoint transformations of $\mathfrak{g}$. This gives us a decomposition of $\mathfrak{g}$ as the orthogonal direct sum of simultaneous eigenspaces. All the eigenvalues will be real. In particular, the simultaneous eigenvalues will be members of the dual space $\mathfrak{a}^*$. If $\lambda \in \mathfrak{a}^*$, we define:

$$\mathfrak{g}_\lambda = \{X \in \mathfrak{g} \mid (\operatorname{ad} H)X = \lambda(H)X \text{ for all } H \in \mathfrak{a}\}$$

If $\mathfrak{g}_\lambda \neq 0$ and $\lambda \neq 0$, λ is a *restricted root* of $\mathfrak{g}$. We let Σ denote the set of restricted roots.

There are a few relevant facts about the maximal abelian subspaces of $\mathfrak{p}$. If $\mathfrak{a}$ and $\mathfrak{a}'$ are both maximal abelian subpaces, then there is an element k of K such that $\operatorname{Ad}(k)\mathfrak{a}' = \mathfrak{a}$. In particular, we have:

$$\mathfrak{p} = \bigcup_{k \in K} \mathrm{Ad}(k)\mathfrak{a}$$

For a set of restricted roots Σ, we can pick a set of *positive roots*, which we denote $\Sigma^+ \subset \Sigma$. The positive roots form a halfspace of Σ. This means that $\Sigma^- := -\Sigma^+$, satisfies $\Sigma = \Sigma^+ \sqcup \Sigma^-$. The roots of Σ^+ that are not a sum of any other roots are called the *simple roots*, and are denoted $\Delta \subset \Sigma^+$.

The results in this section apply to a more general class of Lie groups, namely reductive Lie groups, but as the focus of this work is on semisimple Lie groups, we will work with those, instead of in full generality.

For a semisimple Lie group G, we let θ be a Cartan involution on the Lie algebra $\mathfrak{g}_0$ of G, and $\mathfrak{g} = \mathfrak{k}_0 \oplus \mathfrak{p}_0$ be the corresponding Cartan decomposition. We choose a maximal abelian subspace $\mathfrak{a}_0$ of $\mathfrak{p}_0$. Now, recall the global Cartan decomposition of G as $G = K \exp \mathfrak{p}_0$. Since $\mathfrak{p}_0 = \bigcup_{k \in K} \mathrm{Ad}(k)\mathfrak{a}_0$, we can see that any $g \in G$ can be written as $g = k \exp \mathrm{Ad}(k')H$ for some $k, k' \in K$ and $H \in \mathfrak{a}_0$. We can then write

$$g = k(k' \exp(H)(k')^{-1}) = (kk') \exp(H)(k')^{-1}$$

From which we see that we can decompose G into $G = KAK$, where A is the subgroup of G with Lie algebra $\mathfrak{a}_0$.

We can now give the uniqueness of the KAK decomposition. In particular, any $g \in G$ can be written as $g = k_1 \exp(H) k_2$. $H \in \mathfrak{a}_0$ is unique, and this defines the *Cartan projection* $\mu : G \to \mathfrak{a}_0$ by $g \in G \mapsto H \in \mathfrak{a}_0$. Then the elements k_1, k_2 are unique up to multiplication by an element of $M = Z_k(\mathfrak{a}_0)$, an element of K that centralizes $\exp(\mathfrak{a}_0^+)$.

2.2 Geometric Structures

We now turn to the topic of geometric structures. Some references for this section include the survey paper by Alessandrini [Ale19] and the book by Goldman [Gol22]. These are generalizations of the idea that a manifold is 'locally Euclidean', to include other geometries than Euclidean. A geometry is defined as a pair (G, X) where G is a Lie group and X is the *model space*, which admits a transitive and effective action of G. Effective means that only the identity acts trivially on X. For any $x \in X$, we define the isotropy group of x in G to the be stabilizer of x:

$$\mathsf{H} = \mathrm{Stab}_{\mathsf{G}}(x) = \{g \in \mathsf{G} \mid g \cdot x = x\}$$

Since the action of G on X is transitive, any pair of $x_1, x_2 \in X$ have isomoprhic stabilizers under conjugation by the element $g \in \mathsf{G}$ such that $g \cdot x_1 = x_2$.

Example 2.5. The classic example is the Euclidean geometry: $(\mathrm{Isom}(\mathbb{R}^n), \mathbb{R}^n)$. Other examples include:

- Hyperbolic geometry: $(\mathsf{PO}(1, n), \mathbb{H}^n)$
- Spherical geometry: $(\mathsf{PO}(n+1), S^n)$

These are all examples of *Riemannian type* geometries as they preserve a Riemannian metric.

Example 2.6. Complex Hyperbolic geometry is the pair $(\mathsf{SU}(n, 1), \mathbb{CH}^n)$.

Definition 2.7. A *geometric structure* on a manifold M, for a given (G, X) pair with $\dim(M) = \dim(X)$, is a maximal atlas $\{(U_i, \varphi_i)\}$ where:

- The collection $\{U_i\}$ forms an open cover of M

- The functions $\varphi_i : U_i \to X$ are homeomorphisms to their image
- The transition functions:

$$\varphi_i \circ \varphi_j^{-1} : \varphi_j(U_i \cap U_j) \longrightarrow \varphi_i(U_i \cap U_j)$$

  are locally in G, which means $(\varphi_i \circ \varphi_j^{-1})|_C = g|_C$ for some $g \in \mathsf{G}$, and for every connected component C of $\varphi_j(U_i \cap U_j)$.

A manifold together with a geometric structure is an (G, X)-*manifold*.

For any simply connected (G, X)-manifold N, there is a map: $\mathrm{Dev} : N \to X$ such that for any $n \in N$, there is chart (U, φ) around n where $\mathrm{Dev} \circ \varphi^{-1}$ is locally in G. This map is unique, up to composition with an element of G. If N is the universal cover of some manifold M, then such a map is called a developing map for M.

Definition 2.8. Let M be a manifold without an (G, X)-structure. A pair (Dev, h) is a *developing pair* for M if:

- h is a representation $h : \pi_1(M) \to \mathsf{G}$
- $\mathrm{Dev} : \widetilde{M} \to X$ is an h-equivariant local diffeomorphism

Given such a developing pair, M can be endowed with a (G, X)-structure in a canonical manner. For a simply connected open subset $U \subset M$, take a section $s : M \to \widetilde{M}$ (assuming that U is small enough so that $s(U)$ is an open subset where Dev is a diffeomoprhism). Then $(U, \mathrm{Dev} \circ s)$ is a chart. The collection of all charts of this type is a maximal atlas for a (G, X)-structure on M.

In Chapters 5 and 6, we shall use Higgs bundles to construct developing pairs for certain manifolds and thus equip them with a geometric structure.

Chapter 3

The Nonabelian Hodge Correspondence and Higgs Bundles

Higgs Bundles provide a core tool for all of our results. In this chapter we will define them and give a rough explanation of how they relate to representations and the character variety. This chapter will forego many details of these relations, which collectively are called the nonabelian Hodge correspondence, as these details are not relevant to this specific thesis.

3.1 Higgs Bundles

Let G be a reductive Lie group with a maximal compact subgroup H. We fix a Cartan decomposition of $\mathrm{Lie}(\mathsf{G}) = \mathfrak{g}$ as $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$. We can complexify the $\mathfrak{g}$ which gives a decomposition as:

$$\mathfrak{g}^{\mathbb{C}} = \mathfrak{h}^{\mathbb{C}} \oplus \mathfrak{m}^{\mathbb{C}}$$

This decomposition is $\mathrm{Ad}_{H^{\mathbb{C}}}$ invariant, a result that derives from the bracketing relations a Cartan decomposition must satisfy: $[\mathfrak{h}, \mathfrak{m}] \subset \mathfrak{m}$ and $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$.

Remark 3.1. We will reserve the symbol K for a separate object in this chapter, so we will adjust the notation for our compact from K to H, leading to a switch in notation convention from the previous chapter, where we used standard Lie theoretic notation.

For a fixed Riemann surface X with genus $g \geq 2$, we will use K to denote it's holomorphic cotangent bundle. Now given a principal $\mathsf{H}^{\mathbb{C}}$-bundle P, we have the associated complex vector bundle $P[\mathfrak{m}^{\mathbb{C}}]$ with fiber $\mathfrak{m}^{\mathbb{C}}$ defined by the equation:

$$P[\mathfrak{m}^{\mathbb{C}}] := (P \times \mathfrak{m}^{\mathbb{C}})/\sim, \qquad \text{where } (p \cdot h, v) \sim (p, \mathrm{Ad}_{h^{-1}} v) \text{ for } h \in \mathsf{H}^{\mathbb{C}}$$

We can now define one of our central objects.

Definition 3.2. A G-Higgs bundle over X is a pair (P, φ) where $P \to X$ is a holomorphic principal $\mathsf{H}^{\mathbb{C}}$- bundle and φ, called the *Higgs field* is a holomorphic section of the associated bundle $P[\mathfrak{m}^{\mathbb{C}}] \otimes K$, i.e. $\varphi \in H^0(X, P[\mathfrak{m}^{\mathbb{C}}] \otimes K)$.

Example 3.3. *Compact groups*: If G is compact, then $\mathsf{H} = \mathsf{G}$, so the Cartan decomposition trivializes as $\mathfrak{g}^{\mathbb{C}} = \mathfrak{h}^{\mathbb{C}} \oplus 0$, where $\mathsf{m}^{\mathbb{C}} = 0$. Then a G-Higgs bundle is simply a holomorphic $G^{\mathbb{C}}$ bundle over X.

Example 3.4.

Complex groups: Let G be a complex Lie group. The only Cartan decompositions of $\mathfrak{g}^{\mathbb{R}}$ are conjugation with respect to a compact real form. Fix a compact real form $\mathfrak{h}$. Then $\mathfrak{g}^{\mathbb{R}} = \mathfrak{h} \oplus i\mathfrak{h}$, where $\mathfrak{h} \cong \mathfrak{m}$. Also, $\mathfrak{g} = \mathfrak{h}^{\mathbb{C}}$. The complexification of the decomposition is $(\mathfrak{g}^{\mathbb{R}})^{\mathbb{C}} = \mathfrak{h}^{\mathbb{C}} \oplus \mathfrak{h}^{\mathbb{C}} \cong \mathfrak{g} \oplus \mathfrak{g}$. So a G-Higgs bundle is a principal G-bundle P, and $\varphi \in H^0(X, P[\mathfrak{g}] \otimes K)$.

It is often more convienent to work with a vector bundle rather than a principal bundle. The following examples will demonstrate how to work with an equivalent vector bundle and how the structure of G gives rise to structures on G-Higgs bundles.

Example 3.5. Let $G = \mathsf{SL}(n, \mathbb{C})$. This is a complex Lie group, so the previous example can be applied: the Lie algebra is $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{C})$, which has compact real form $\mathfrak{su}(n)$. So a $\mathsf{SL}(n, \mathbb{C})$-Higgs bundle is a principal $\mathsf{SL}(n, \mathbb{C})$-bundle P with Higgs field $\varphi \in H^0(X, P[\mathfrak{sl}(n, \mathbb{C})] \otimes K)$.

Now, we will take the standard representation, which is faithful, of $\mathsf{SL}(n, \mathbb{C}) \to \mathsf{GL}(\mathbb{C}^n)$. This also defines an embedding of $\mathfrak{sl}(n, \mathbb{C})$ into $\operatorname{End}(\mathbb{C}^n)$. Then we can construct the associated vector bundle to P with fiber $\mathbb{C}^n$ using the representation.

$$P[\mathbb{C}^n] := P \times C^n / \sim$$

where $\sim$ is the equivalence relation $(x, v) \sim (x \cdot g, g \cdot v)$ for $(x, v) \in P \times C^n$. The right action on x is the action of the principal bundle, and the left action on v is given by the standard representation.

Let $E = P[\mathbb{C}^n]$ be the associated bundle just described. Te embedding of $\mathfrak{sl}(n, \mathbb{C})$ into $\operatorname{End}(\mathbb{C}^n)$ lets us define $\Phi \in H^0(X, \operatorname{End}(E) \otimes K)$, which we will call the Higgs field when we are working with the holomorphic vector bundle E. Since we have the standard embedding of $\mathfrak{sl}(n, \mathbb{C})$ into the endomorphisms, $\operatorname{Tr} \Phi = 0$. We also know that the standard representation of $\mathsf{SL}(n, \mathbb{C})$ preserves the standard volume form on $\mathbb{C}^n$, so E comes with a holomorphic volume form $\omega \in H^0(\Lambda^n E)$. The existence of this volume form is equivalent to $\Lambda^n E$ being the trivial line bundle. So a $\mathsf{SL}(n,\mathbb{C})$-Higgs bundle is equivalent to a pair (E,Φ), where E is a holomorphic vector bundle satisfying $\Lambda^n E = \mathcal{O}$ and $\Phi \in H^0(X, \mathrm{End}(E)\otimes K)$. We will often refer to this as the Higgs bundle.

This example gives rise to the general process we will use in order to work with vector bundle formulations of Higgs bundles. Suppose we have a G-Higgs bundle P with Higgs field φ. We start with a faithful representation from $\mathsf{G}^{\mathbb{C}} \to \mathsf{GL}(V)$, which will induce a representation of $\mathsf{H}^{\mathbb{C}} \to \mathsf{GL}(V)$ and an embedding of $\mathfrak{m}^{\mathbb{C}} \to \mathrm{End}(V)$. The action of $\mathsf{H}^{\mathbb{C}}$ on V given by the representation lets us construct an associated vector bundle with fiber V to the principal $\mathsf{H}^{\mathbb{C}}$-bundle P, which we will write as $E = P[V]$. Then the Higgs field $\varphi \in H^0(X, P[\mathfrak{m}^{\mathbb{C}}]\otimes K)$ can be embedded into $H^0(X, \mathrm{End}(E)\otimes K)$ and we will denote it as Φ. This follows from the chain of inclusions $\mathfrak{m}^{\mathbb{C}} \to \mathrm{End}(V)$, which induces $P[\mathfrak{m}^{\mathbb{C}}]\otimes K \to \mathrm{End}(P[V])\otimes K$, where the last term is, of course, equivalent to $\mathrm{End}(E)\otimes K$.

Example 3.6. Let $G = \mathsf{SL}(n,\mathbb{R})$. The maximal compact is $\mathsf{SO}_0(n)$. Further, if we take the Cartan involution to be the negative conjugate transpose (which simplifies to negative transpose since our matrices have real entries), the $+1$-eigenspace of the Cartan involution is $\mathfrak{so}(n)$, so Cartan decomposition agrees with this maximal compact. Then -1-eigenspace will be traceless symmetric matrices, which we denote as $\mathrm{Sym}_0(\mathbb{R}^n)$. The Cartan decomposition is:

$$\mathfrak{sl}(n,\mathbb{R}) = \mathfrak{so}(n) \oplus \mathrm{Sym}_0(\mathbb{R}^n)$$

We again use the standard representation of $\mathsf{G}^{\mathbb{C}} = \mathsf{SL}(n,\mathbb{C})$, so as in the previous example, we have a pair (E,Φ), where E is a holomorphic vector bundle satisfying $\Lambda^n = \mathcal{O}$ and $\Phi \in H^0(X, \mathrm{End}(E)\otimes K)$.

We have further structure however, given by the fact that $H^{\mathbb{C}} = \mathsf{SO}_0(n, \mathbb{C})$, so there is a reduction of structure group which preserves a symmetric complex bilinear form on $\mathbb{C}^n$. Hence we have a form $Q \in H^0(X, S^2(E) \otimes K)$ which is everywhere non-degenerate. As $\mathfrak{m}^{\mathbb{C}} = \mathrm{Sym}_0(\mathbb{C}^n)$, the Higgs field Φ is symmetric with respect to Q: $\Phi^T Q = Q\Phi$.

Thus an $\mathsf{SL}(n, \mathbb{R})$ Higgs bundle is given by the data of (E, Q, Φ) as described above.

Example 3.7. Let $G = \mathsf{SO}_0(p, q)$. The maximal compact subgroup H is isomorphic to $S(O(p) \times O(q))$, so $\mathfrak{h} = \mathfrak{so}(p) \oplus \mathfrak{so}(q)$. We can decompose $\mathbb{R}^{p+q} = \mathbb{R}^p \oplus \mathbb{R}^q$. Then we can decompose a matrix $X \in \mathrm{End}(\mathbb{R}^{p+q})$ into block form as $X = \left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right]$ where $A \in \mathrm{End}(\mathbb{R}^p)$, $D \in \mathrm{End}(\mathbb{R}^q)$ and $B \in \hom(\mathbb{R}^p, \mathbb{R}^q)$, $C \in \hom(\mathbb{R}^q, \mathbb{R}^p)$. In this language, we write:

$$\mathfrak{sl}(p, q) = \left\{ \begin{bmatrix} A & B \\ C & D \end{bmatrix} \middle| \begin{bmatrix} A & B \\ C & D \end{bmatrix}^T \begin{bmatrix} \mathrm{Id}_p & 0 \\ 0 & -\mathrm{Id}_q \end{bmatrix} + \begin{bmatrix} \mathrm{Id}_p & 0 \\ 0 & -\mathrm{Id}_q \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = 0 \right\}$$

From this relation, we see that an $X = \left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right] \in \mathfrak{so}(p+q)$ must satisfy $A \in \mathfrak{so}(p)$, $D \in \mathfrak{so}(q)$ and $B = -C^T$. From this, the Cartan decomposition is clearly:

$$\mathfrak{so}(p, q) = (\mathfrak{so}(p) \oplus \mathfrak{so}(q)) \oplus \hom(\mathbb{R}^p, \mathbb{R}^q)$$

There is a natural representation of $\mathsf{H}^{\mathbb{C}} = S(O(p, \mathbb{C}) \times O(q, \mathbb{C}))$ into the groups: $\mathsf{SL}(n, \mathbb{C})$, $\mathsf{SO}_0(p+q, \mathbb{C})$ and $\mathsf{S}(\mathsf{O}(p) \times \mathsf{O}(q))$, with each level further restricting the representation. So we get all the structures, each one refining to the next. We know we have a vector bundle E with fibers $\mathbb{C}^{p+q}$ satisfying $\Lambda^n E = \mathcal{O}$, and a symmetric bilinear form on each fiber. The final restriction to $\mathsf{H}^{\mathbb{C}}$ says that there is an orthogonal splitting with respect to Q on the fibers of E into $\mathbb{C}^{p+q} = \mathbb{C}^p \oplus \mathbb{C}^q$. Thus, we can write $E = V \oplus W$ where V is rank p and W

is rank q, and $Q = \begin{bmatrix} Q_V & 0 \\ 0 & Q_w \end{bmatrix}$ where Q_V and Q_W are symmetric bilinear forms on V and W respectively. We define a signature (p,q) form on $E = V \oplus W$ by the form $Q_V \oplus -Q_W$.

Since $\mathfrak{m}^{\mathbb{C}} = \hom(\mathbb{C}^p, \mathbb{C}^q)$, the Higgs field satisfies $\Phi \in H^0(X, \mathrm{End}(V \oplus W) \otimes K)$:

$$\Phi = \begin{bmatrix} 0 & \eta^\dagger \\ \eta & 0 \end{bmatrix}$$

where $\eta \in H^0(X, \hom(V,W) \otimes K)$. Using the identification of Q_V with an isomorphism $V \to V^*$ and Q_W with $W \to W^*$, we see that $\eta^\dagger = -Q_V^{-1} \eta^T Q_W$.

Definition 3.8. An $\mathsf{SO}_0(p,q)$-*Higgs bundle* on X is a tuple (V, W, Q_V, Q_W, η) satisfying:

- V and W are respectively rank p and q holomorphic vector bundles over X satisfying $\Lambda^{p+q}(V \oplus W) = \mathcal{O}$, i.e. $\Lambda^p V \cong \Lambda^q W^*$.
- $Q_V : V \to V^*$ and $Q_W : W \to W^*$ are holomorphic symmetric isomorphisms.
- A holomorphic section $\eta \in H^0(X, \hom(V,W) \otimes K)$.

Example 3.9. Let $G = \mathsf{SU}(p,q)$. The maximal compact subgroup is $\mathsf{H} = \mathsf{S}(\mathsf{U}(p) \times \mathsf{U}(q))$, so

$$\mathfrak{h}^{\mathbb{C}} = \mathfrak{sl}(p,\mathbb{C}) \times \mathfrak{sl}(q,\mathbb{C}) \subset \mathfrak{sl}(p+q,\mathbb{C})$$

Since $\mathfrak{g}^{\mathbb{C}} = \mathfrak{sl}(p+q,\mathbb{C})$, we can identify $\mathfrak{m}^{\mathbb{C}}$ of the complexified Cartan decomposition with the off diagonal block matrices in $\mathfrak{sl}(p+q,\mathbb{C})$. *A fortiori*, a $\mathsf{SU}(p,q)$-Higgs bundle is an $\mathsf{SL}(p+q,\mathbb{C})$-Higgs bundle, so we have the data (E,Φ) for a rank $p+q$ holomorphic vector bundle E such that $\Lambda^n E = \mathcal{O}$ and $\Phi \in H^0(X, \mathrm{End}(E) \otimes K)$. The reduction of structure from $\mathsf{SL}(p+q,\mathbb{C})$ to $\mathsf{SU}(p,q)$ induced a reduction on the principal Higgs bundle from $\mathsf{SL}(p+q,\mathbb{C})$, to $\mathsf{H}^{\mathbb{C}} = (S)(\mathsf{GL}(p,\mathbb{C}) \times \mathsf{GL}(q,\mathbb{C}))$. This induces a splitting of $E = V \times W$ where V is a rank p vector bundle, and W is a rank q vector bundle. It also ensures that the Higgs field has the form:

$$\Phi = \begin{bmatrix} 0 & \beta \\ \gamma & 0 \end{bmatrix}$$

where $\beta : W \to V \otimes K$ and $\gamma : V \to W \otimes K$. Thus we can make the following definition:

Definition 3.10. A $\mathsf{SU}(p,q)$-Higgs bundle on X is a tuple (V, W, β, γ) such that:

- V and W are respectively rank p and q holomorphic vector bundles satisfying $\Lambda^{p+q}(V \oplus W) = \mathcal{O}$, i.e. $\Lambda^p V \cong \Lambda^q W^*$.
- $\beta \in H^0(X, \mathrm{hom}(W,V) \otimes K)$ and $\gamma \in H^0(X, \mathrm{hom}(V,W) \otimes K)$.

3.2 Stability and Harmonics Metrics

The main results we prove will be for $\mathsf{SU}(n+1,1)-$ and $\mathsf{SO}_0(2,n+2)-$Higgs bundles. In order to prove them, we will need a Hermitian metric for which the splittings of Definitions 3.10 and 3.8 are orthogonal. It turns out that the existence of such a metric is guaranteed if the bundles are polystable.

Underlying both families of Higgs bundles is an $\mathsf{SL}(m,\mathbb{C})$-Higgs bundle. It will suffice that the underlying $\mathsf{SL}(m,\mathbb{C})$-Higgs bundles are stable.

Definition 3.11. Let (E, Φ) be an $\mathsf{SL}(m,\mathbb{C})$-Higgs bundle.

- The bundle is *stable* if all Φ invariant subbundles, that is, $F \subset E$ with $\Phi(F) \subset F \otimes K$, have $\deg(F) < 0$.

- The bundle is *polystable* if it is a direct sum of stable bundles, $E = \bigoplus E_j$ where $\deg(E_j) = 0$ and each E_j is stable as an $\mathsf{SL}(m_j, \mathbb{C})$-Higgs bundle.

An $\mathsf{SU}(n+1,1)$-Higgs bundle is polystable if and only if it's underlying $\mathsf{SL}(n+2,\mathbb{C})$-Higgs bundle is polystable. Similarly, an $\mathsf{SO}_0(2, n+2)$-Higgs bundle is polystable if and only if it's corresponding $\mathsf{SL}(n+4, \mathbb{C})$-Higgs bundle is polystable.

There is a general theory relating stability to the existence of Hermitian metrics which solve Hitchins equations [Hit87a, Sim88, GPGMiR09]. In this work, we will only use two special cases of the general theory for reductive Lie groups explored in [GPGMiR09] . First, we state the result for $\mathsf{SU}(n+1,1)$-Higgs bundles from [BGPG03].

Theorem 3.12 ([BGPG03])**.** *Let (V, W, β, γ) be an $\mathsf{SU}(n+1,1)$-Higgs bundle. Then it is polystable if and only if there exists a hermitian metric such that $E = V \oplus W$ is an orthogonal decomposition with respect to the metric and such that Hitchin's equations are satisfied.*

We shall not concern ourselves with the details of Hitchin's equations. The relevant implication to our work is that, for the metric h guaranteed by the theorem, there is an associated Chern connection on E, ∇^{Ch}. Using the Chern connection and the data of the Higgs field, we construct the connection:

$$\nabla = \nabla^{Ch} + \Phi + \Phi^* \tag{3.1}$$

If h satisfies the Hitchin equations, then ∇ is a flat connection on E, where the adjoint of Φ is taken with respect to h. We shall refer to h as either the harmonic metric, or the positive definite hermitian metric. So polystability of a bundle endows it with this Harmonic metric and a flat connection.

The orthogonality of the splitting with respect to h means that h decompses into $h = h_V \oplus h_W$, Hermitian metrics on V and W. We then have a signature $(n+1, 1)$ indefinite Hermitian metric on E given by $h_I = h_V \oplus -h_W$.

There is a similar result for $\mathsf{SO}_0(2, n+2)$-Higgs bundles. We use the formulation given in [CTT19]. First, we introduce the idea of an adapted Hermitian metric. A Hermitian metric h on E is *adapted* to a $\mathbb{C}$-bilinear symmetric form Q if:

$$h(u, v) = Q(u, \lambda(v))$$

for $\lambda : E \to E$ an anti-linear involution.

Theorem 3.13 ([CTT19])**.** *Let (V, W, Q_V, Q_W, η) be a $\mathsf{SO}_0(2, n+2)$-Higgs bundle. The bundle is polystable if and only if there exists Hermitian metrics h_V and h_W on V and W, adapted to Q_V and Q_W respectively, such that $h = h_V \oplus h_W$ satisfies the Hitchin equations on $E = V \oplus W$.*

As in the case of $\mathsf{SU}(n+1, 1)$, for the purpose of this thesis, satisfying the Hitchin equations means that the connection given by Equation 3.1 is flat. Since the metrics h_V and h_W are adapted, they correspond to $h_V(\cdot, \lambda_V(\cdot)) = Q_V$ and $h_W(\cdot, \lambda_W(\cdot)) = Q_W$ and $Q_V \oplus -Q_W$ restricted to the associated real bundle $E_{\mathbb{R}}$ is a ∇-parallel signature $(2, n+2)$ metric on $E_{\mathbb{R}}$.

We will now establish some general properties and conventions regarding metrics on Higgs bundles. As far as we are concerned, all the Higgs bundles we work with will be bundles over a compact Riemann surface equipped with a positive definite Hermitian metric h, and also an indefinite metric h_I. We fix a basepoint $x_0 \in X$ and choose a maximal compact

$\mathsf{H}(x_0)$ corresponding to the metric h at x_0. Then the metric on π^*E is equivalent to a $\pi_1(X)$-equivariant map:

$$\widetilde{h} : \widetilde{X} \to \mathsf{G}/\mathsf{H}(x_0)$$

where G is the holonomy group of E. That is, $\widetilde{h}$ assigns to each point in $\widetilde{H}$ an element of the Riemannian symmetric space $\mathsf{G}/\mathsf{H}(x_0)$.

One thing we will often do is pullback our bundles to bundles over $\widetilde{X}$. Let $\pi : \widetilde{X} \to X$ be the normal covering map, and $p : E \to X$ be the bundle projection. Then the pullback π^*E over $\widetilde{X}$ is defined as:

$$\pi^*E = \{(x, e) \in \widetilde{X} \times E \mid \pi(x) = p(e)\}$$

As the $\widetilde{X}$ is connected and simply connected, and π^*E is a flat bundle, there is a trivialization of π^*E as $\widetilde{X} \times E(x_0)$, where $E(x_0)$ is the fiber of E over x_0. This trivialization is given by choosing a frame in the fiber over a basepoint and parallel transporting it to every fiber, which gives a basis of global sections.

Using this trivialization of π^*E, the metric at a point $x \in \widetilde{X}$ for $u, v \in E(x_0)$ is given by:

$$h(u, v)_x := h(\widetilde{h}(x)^{-1}u, \widetilde{h}(x)^{-1}v)_{x_0}$$

where $\widetilde{h}(x)$ can be any coset representive. In particular, for $\|v\|_h(x)$, the norm of a vector v with respect to the metric h at x, is computed with the formula

$$\|v\|_{h(x)} = \|\widetilde{h}(x)^{-1}v\|_{h(x_0)} \tag{3.2}$$

Also, with this definition, we see that the norm is left $\pi_1(X)$-invariant:

$$
\begin{aligned}
\|\rho(\gamma)v\|_{h(\gamma\cdot x)} &= \|\widetilde{h}(\gamma\cdot x)^{-1}\rho(\gamma)v\|_{h(x_0)} \\
&= \|\left(\rho(\gamma)\widetilde{h}(x)\right)^{-1}\rho(\gamma)v\|_{h(x_0)} \\
&= \|\widetilde{h}(x)^{-1}\rho(\gamma)^{-1}\rho(\gamma)v\|_{h(x_0)} \\
&= \|\widetilde{h}(x)^{-1}v\|_{h(x_0)} \\
&= \|v\|_{h(x)}
\end{aligned}
\tag{3.3}
$$

3.3 The Correspondence Between Higgs Bundles and Representations

The core theme of this thesis is studying representations and the character variety using Higgs bundles. So we need a way to go back and forth between the two objects in order to understand how properties of one affect the other.

Given a polystable G-Higgs bundle for $\mathsf{G} = \mathsf{SU}(n+1,1)$ or $\mathsf{SO}(2,n+2)$, the existence of the harmonic metric from Theorems 3.12 and 3.13 induced a flat structure on the bundle. This then gives the holonomy representation $\rho : \pi_1(X) \to \mathsf{G}$. This representation is completely reducible.

One the other hand, given a representation $\rho : \pi_1(X) \to \mathsf{G}$, we can define a holomorphic vector bundle with fiber V by:

$$E_\rho := \widetilde{X} \times_{\pi_1(X)} V$$

where $\pi_1(X)$ acts on $\widetilde{X}$ by deck transformations, and on V on the left by $\rho(\gamma)$. As the transition maps are locally constant, E_ρ naturally has a flat connection. Moreover, if the representation is reductive, the E_ρ admits a harmonic metric as a result of the Corlette-Donaldson theorem [Cor88, Don87]. Then the flat connection decomposes into a Chern connection with respect to the metric and a $(1,1)$-form, the $(1,0)$ part corresponds to the Higgs field data.

This extends to a famous results know as the nonabelian Hodge correspondence, which is a homeomorphism between the character variety, also known as the Betti moduli space:

$$\mathcal{M}^B(\Sigma, \mathsf{G}) := \hom^{\mathrm{Red}}(\pi_1(\Sigma), \mathsf{G})/\mathsf{G}$$

that is, the moduli space of reductive representations modulo a G-action: $(g\cdot\rho)(\gamma) = g^{-1}\rho(\gamma)g$, and the Dolbeault moduli space

$$\mathcal{M}^{\mathrm{Dol}}(X, \mathsf{G})$$

of isomorphism classes of polystable G-Higgs bundles. The details of this are outside the scope of this thesis, so it suffices to accept that there is a relation between these objects, and we shall see how certain structures of Higgs bundles correspond to features of their associated represenations.

3.4 Toledo Invariants

An important invariant of representations in the character variety is the Toledo invariant. We will give the definition used in [BGPG03] for $\mathsf{SU}(p,q)$ representations. This definition is given in terms of the Higgs bundle (V, W, β, γ) corresponding to a given representation $\rho : \pi_1(X) \to \mathsf{SU}(p,q)$. Let $a = \deg V$ and $b = \deg W$. Then the Toldeo invariant τ is given by the formula:

$$\tau = 2\frac{aq - bp}{p + q}$$

In particular, for $p = 2$ and $q = 1$, we know that $\deg V + \deg W = 0$, so $b = -a$ and the equation simplifies to:

$$\tau = 2\frac{a - (-a)2}{3} = 2\frac{3a}{3} = 2a$$

And we see that τ is an integer. There is an inequality known as the Milnor-Wood inequality which gives a bound on τ. For the $\mathsf{SU}(2,1)$, this bound is

$$|\tau| \leq 2g - 2$$

So the possible values of the Toledo invariant for $\mathsf{SU}(2,1)$ are even integers in $[2 - 2g, 2g - 2]$. A result of Xia shows that the Toledo invariant differentiates connected components of $\mathcal{M}^B(\Sigma, \mathsf{SU}(2,1))$ and conversely, that there is a component of the character variety corresponding to each allowable τ [Xia03].

3.5 Anosov Representations

One property a representation may have is the Anosov property. This is a generalization of the notion of convex cocompact representations. Convex cocompact representations in rank 1 Lie groups were of interest since representations in the Hitchin component of $\mathsf{PSL}(2, \mathbb{R})$ were discrete and faithful, and convex cocompact guarantees those properties. In the search for other components consisting of discrete and faithful representations, convex cocompactness was thus a useful tool. As mentioned in the introduction, generalizing the definition in a straightforward way yielded no new representations [KL05]. A search for an appropriate generalization was started and Labourie found a dynamical property which he called the *Anosov condition* [Lab06]. Importantly, any representation that is Anosov will also be discrete and faithful. This definition was generalized further, and in particular, by Guichard and Weinhard [GW12] and Kapovich, Leeb and Porti [KLP17]. The definition we shall use is found in Filip [Fil22] and is a specialization of the definitions found in the above references.

We will let G be a reductive Lie group with $\theta \subset \Delta$ a subset of simple roots and, fix a positive Weyl chamber $\mathfrak{a}^+$. Let X be a compact Riemann surface of genus $g \geq 2$. Let $x_0 \in \widetilde{X} \cong \mathbb{H}^2$ be a fixed point and use the hyperbolic metric on $\mathbb{H}^2$ to define the distance $d(x, y)$ between points $x, y \in \widetilde{X}$. We let $\mu : \mathsf{G} \to \mathfrak{a}^+$ be the Cartan projection.

Definition 3.14. A representation $\rho : \pi_1(X) \to \mathsf{G}$ is θ-Anosov if there exists $C_1, C_2 > 0$ such that

$$\alpha(\mu(\rho(\gamma))) \geq \frac{1}{C_1} d(x_0, \gamma x_0) - C_2$$

for all $\gamma \in \pi_1(X)$ and $\alpha \in \theta$.

Associated to an Anosov representation is a *domain of discontinuity*, that is, an open subset Ω of a compact G-space where:

- The action of ρ on Ω fixes Ω
- The action of ρ is properly discontinuous and the quotient Ω/ρ is compact.

Chapter 4

Gradient Estimates and Exponential Growth

4.1 Gradient Estimates

In this section, we will focus on an a class of abstract functions and derive some key properties of their gradients. This is based on [Fil22] but generalized so that we can construct functions on several different classes of Higgs bundles. Further background comes from [Str96, Bis15]. We will start by recalling some tools necessary for the following proofs.

4.1.1 Preliminary Details

Our ultimate goal is to prove that the functions of interest have unique, non-degenerate critical points. In order to prove such results, it is necessary to have to certain conditions to avoid pathological cases where a function's behavior is 'bad'. We start with the following definition. We shall work on a complete Riemannian manifold M, with metric g, and $\|\cdot\|_g$

will denote the norm associated to g and the gradient ∇F will be understood to be the gradient associated to the Riemannian metric. We will denote the Riemannian distance function as $d : M \times M \to \mathbb{R}^+$.

Definition 4.1. Let $F : M \to R$ a continuously differentiable function. A *Palais-Smale* sequence for F is a sequence of points $\{x_n\}_{n\in\mathbb{N}}$ such that

- The sequence $|F(x_n)|$ is bounded and
- $\|df(x_n)\|_g \to 0$

We will also denote such a sequence as a *(PS)-sequence.*

A sign of the 'bad' behavior we wish to avoid is the existence of a (PS)-sequence which does not converge. An example due to [Bis15] of such bad behavior is simple. Consider the polynomial $F(x,y) = (x^2y - x - 1)^2 + (x^2 - 1)^2$. The sequence $(x_n, y_n) = \left(\frac{1}{2n} + \frac{1}{2n^2}, n + \frac{1}{n}\right)$ is a (PS)-sequence that does not converge.

In order to avoid such behavior, we will focus of functions which satsify the the *Palais-Smale condition* (or (PS)-condition for short), which requires every (PS)-sequence to have a convergent subsequence.

To start, we will consider functions $F : M \to R$ that are twice differentiable and satisfy the following three conditions:

(F1) $F(x_0) = 0$ for some $x_0 \in M$.

(F2) There is an $r > 0$ and an $\alpha > 0$ such that $F(x) \geq \alpha$ for all x with $d(x, x_0) = r$.

(F3) There is an x_1 such that $d(x_0, x_1) > r$ and $F(x_1) < \alpha$.

In [Bis15, Theorem 3.1], Bisgard proved a result for functions in Euclidean space that we will generalize to manifolds.

Theorem 4.2. *Any twice differentiable function $F : M \to R$ which satisfies (F1)-(F3) has a (PS)-sequence x_n such that $F(x_n) \to c$ for some $c \geq \alpha$.*

The key idea of this proof is that given any path γ connecting x_0 and x_1 from condition (F3), there must be points $x' \in \gamma$ such that $F(x') \geq \alpha$ by condition (F2). If we then move all the points in γ 'downhill' relative to F, the new path must still connect x_0 and x_1, so there must be points that are not moved below the level α. Since moving downhill corresponds to moving in the negative gradient direction, the fact that these points move very little indicates the length of the gradient is small.

Proof. Our first step is to normalize ∇F, to control its magnitude so it never grows uncontrollably. We define a simple normalization term

$$w(x) = \frac{\|\nabla F(x)\|_g}{1 + \|\nabla F(x)\|_g^2}$$

and now we consider the scaled gradient flow φ_t defined by the equations

$$\frac{d}{dt}\varphi_t(x) = -w(\varphi_t(x))\nabla F(\varphi_t(x))$$

$$\varphi_0(x) = x$$

Now we first check that the norm of $\frac{d}{dt}\varphi_t$ is bounded to guarantee solutions exist for all time t:

$$\begin{aligned}\|w(\varphi_t(x))\nabla F(\varphi_t(x))\|_g &= \frac{\|\nabla F(x)\|_g}{1+\|\nabla F(x)\|_g^2}\|\nabla F(\varphi_t(x))\|_g \\ &= \frac{\|\nabla F(x)\|_g^2}{1+\|\nabla F(x)\|_g^2} \leq 1\end{aligned}$$

Now, we establish that F does indeed decrease as we flow along φ_t forwards in time.

$$\begin{aligned}\frac{d}{dt}F(\varphi_t(x)) &= dF(-w(\varphi_t(x))\nabla F(\varphi_t(x))) \\ &= \langle \nabla F(\varphi_t(x)), -w(\varphi_t(x))\nabla F(\varphi_t(x))\rangle_g \\ &= -w(\varphi_t(x))\langle \nabla F(\varphi_t(x)), \nabla F(\varphi_t(x))\rangle_g \\ &= -w(\varphi_t(x))\|\nabla F(\varphi_t(x))\|_g^2 \leq 0\end{aligned} \tag{4.1}$$

where made use of the relation between the differential of F and the gradient of F.

The next step is to verify that $d(\varphi_t(x_0), x_0) < r$ and $d(\varphi_t(x_1), x_0)$ for all positive times t. This is intuitively clear, since we established that φ_t flows points to lower values of F, and x_0 and x_1 both start at lower values than any point which is distance r away from x_0. To fully justify these inequalities, we will assume for the sake of contradiction that there is a $t > 0$ such that $d(\varphi_t(x_0), x_0) = r$. By Equation 4.1, we know that $F(\varphi_t(x_0)) \leq F(x_0)$. From (F1) and (F2), $F(x_0) < \alpha$ and because $d(\varphi_t(x_0), x_0) = r$, we also know that $\alpha \leq F(\varphi_t(x_0))$. So together, we get:

$$\alpha \leq F(\varphi_t(x_0)) \leq F(x_0) < \alpha$$

which is a clear contradiction, so the first inequality has now been shown.

We then assume that there is some time $t > 0$ such that $d(\varphi_t(x_1), x_0) = r$. By (F3), $F(x_1) < \alpha$, and by Equation 4.1, $F(\varphi_t(x_1)) \leq F(x_1)$, and by (F2), $F(\varphi_t(x_1)) \geq \alpha$ since

$d(\varphi_t(x_1), x_0) = r$. Once, again, these inequalities combine to form a contradiction:

$$\alpha \le F(\varphi_t(x_1)) \le F(x_1) < \alpha$$

Our next step is to join x_0 and x_1 together with a path, and deform that path with φ_t in order to flow the points to lower values. In this process, $\varphi_t(x_0)$ and $\varphi_t(x_1)$ must remain connected and from the previous argument, $\varphi_t(x_0)$ remains with distance r of x_0, while $\varphi_t(x_1)$ stays further than distance r away from x_0. So for all deformations, the path must pass through a point that is precisely distance r away from x_0, hence F applied to that point will be greater than α. We will use this fact to construct a (PS)-sequence.

Let $\gamma : I \to M$ be a continuous path such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. For any $i \in \mathbb{N}$, define the path $\gamma_i : I \to M$ defined by equation $\gamma_i = \varphi_i \circ \gamma$. As γ is continuous, and φ_i is continuous for each i, the path γ_i is continuous for each i. Hence, the image of the compact interval I is compact in M and so by the extreme value theorem, there must be some $s_i \in I$ such that $\gamma_i(s_i)$ is the global maximum. Moreover, we know that

$$d(\gamma_i(0), x_0) = d(\varphi_i(\gamma(0)), x_0) = d(\varphi_i(x_0), x_0) < r$$

$$d(\gamma_i(1), x_0) = d(\varphi_i(\gamma(1)), x_0) = d(\varphi_i(x_1), x_0) > r$$

so by the intermediate value theorem, there must exist an $s' \in I$ such that $d(\gamma_i(s'), x_0) = r$. By (F2), $\alpha \le F(\gamma_i(s'))$. So we know:

$$\alpha \le F(\gamma_i(s')) \le F(\gamma_i(s_i)) \tag{4.2}$$

The sequence $\{s_i\}$ is entirely contained in I, a compact set, so there exists some subsequence $\{s_{i_j}\}$ which converges to $s^* \in I$. Let $x^* = \gamma(s^*)$.

Next we will show that $F(\varphi_t(x^*)) \geq \alpha$ for all $t \geq 0$. Once again, we will use contradiction, so we assume that there is some $t' > 0$ such that $F(\varphi_{t'}(x^*)) < \alpha$. From this and the facts F is continuous and x^* is the limit of the sequence $\{\gamma(s_{i_j}\}$, we know that for all j sufficiently large, we see that

$$F(\varphi_{t'}(\gamma(s_{i_j}))) < \alpha$$

Let J be large enough that $i_J > t'$. By Equation 4.2, we know that $\alpha \leq F(\varphi_{i_J}(\gamma(s_{i_J})))$. From Equation 4.1, since $i_J > t'$, we know that $F(\varphi_{i_J}(\gamma(s_{i_J}))) \leq F(\varphi_{t'}(\gamma(s_{t'})))$. Taken together, we get the following contradiction:

$$\alpha \leq F(\varphi_{i_J}(\gamma(s_{i_J}))) \leq F(\varphi_{t'}(\gamma(s_{t'}))) < \alpha$$

where the last inequality was the consequence of the continuity of F. Therefore, $F(\varphi_t(x^*)) \geq \alpha$ for all $t \geq 0$.

Next, we want estimate the gradient to meet the second condition of a (PS)-sequence. From the previous paragraph, we know that $F(\varphi_t(x^*))$ is bounded below and decreasing as a function of t. From the first fundamental theorem of calculus:

$$\int_0^\infty \frac{d}{dt}F(\varphi_t(x^*))dt = \left(\lim_{t\to\infty} F(\varphi_t(x^*)\right) - F(x^*)$$

which guarantees that

$$\int_0^\infty -\frac{d}{d(}t)F(\varphi_t(x^*))dt < \infty$$

So by Equation 4.2, we see:

$$\begin{aligned}\int_0^\infty w(\varphi_t(x^*))\|\nabla F(\varphi_t(x^*))\|_g^2 dt &= \int_0^\infty -\langle \nabla F(\varphi_t(x^*)), -w(\varphi_t(x^*))\nabla F(\varphi_t(x^*))\rangle_g dt \\ &= \int_0^\infty -dF(-w(\varphi_t(x^*))\nabla F(\varphi_t(x^a st)))dt \\ &= \int_0^\infty -\frac{d}{dt}F(\varphi_t(x^*))dt < \infty\end{aligned}$$

Since this integral is finite, there exists a sequence $\{t_n\}$ such that $t_n \to \infty$ and

$$w(\varphi_{t_n}(x^*))\|\nabla F(\varphi_{t_n}(x^*))\|_g^2 \to 0$$

When we unwrap $w(\varphi_{t_n}(x^*)$, we get:

$$\frac{\|\nabla F(\varphi_{t_n}(x^*))\|_g^3}{1 + \|\nabla F(\varphi_{t_n}(x^*))\|_g^2} \to 0$$

as $n \to \infty$. This implies that $\|\nabla F(\varphi_{t_n}(x^*))\|_g \to 0$. Since $\|\nabla F\| = \|dF\|$, we conclude that $\|dF(\varphi_{t_n}(x^*))\|_g \to 0$ as $n \to \infty$. We already established that $F(\varphi_t(x^*)) \geq \alpha$ for all t, and since $F(\varphi_t(x^*))$ is decreasing, $|F(\varphi_{t_n}(x^*))|$ is bounded. Thus, $x_n := \varphi_{t_n}(x^*)$ is a (PS)-sequence satisfying the conditions of the theorem when $c := \lim_{t\to\infty} F(\varphi_t(x^*))$. $\square$

Another result we need is the Ekeland varational principle [Str96, Theorem 5.1], in the following form:

Theorem 4.3. *Let M be a complete metric space with metric d, and let $F : M \to \mathbb{R} \cup +\infty$ be lower semi-continuous, bounded from below and not identically equal to $+\infty$. Then for any $\epsilon, \delta > 0$ and any $y \in M$ with $F(y) \leq \inf_M F + \epsilon$, there is an element $x \in M$ strictly* *minimizing the functional*

$$F_x(w) := F(w) + \frac{\epsilon}{\delta} d(x, w).$$

Moreover, we have $F(x) \leq F(y)$ *and* $d(x, y) < \delta$*.*

We will not reproduce a proof of this result here. The Ekeland variational principle will allow us to construct a sequence which converges to the infimum of the function while the gradient simulataneously goes to zero.

Proposition 4.4. *Suppose we have a complete Riemannian manifold* M*, and a continuously differentiable function* $F : M \to \mathbb{R}$ *such that* $F^* = \inf_M F \geq 0$*. For every sequence* $\{y_n\} \subset M$ *such that* $F(y_n) \to F^*$ *as* $n \to \infty$*, there exists a second sequence* $\{x_n\} \subset M$ *satsifying:*

(i) $F(x_n) \to F^*$ *as* $n \to \infty$*,*

(ii) $\|\nabla F(x_n)\|_g \to 0$ *as* $n \to \infty$

(iii) $d(x_n, y_n) \to 0$ *as* $n \to \infty$*.*

Proof. Take a sequence $\{y_n\}$ in M such that $F(y_n) \to F^*$. For each n, define $\epsilon_n := F^* - F(y_n)$ and $\delta_n := \sqrt{\epsilon_n}$. We will use these to construct the desired sequence $\{x_n\}$. If $\epsilon_n = 0$, we set $x_n = y_n$. If $\epsilon_n > 0$, then in particular $F(y_n) = F^* - \epsilon_n$, so the Ekeland variational principle supplies an $x_n \in M$ satisfying:

$$F(x_n) \leq F(y_n) \text{ and } d(x_n, y_n) \leq \epsilon_n \tag{4.3}$$

Since x_n also minimizes the functional F_{x_n} from the theorem, we know that for any $z \in M \setminus x_n$, $F_{x_n}(x_n) < F_{x_n}(z)$. Since $\epsilon/\delta = \sqrt{\epsilon}$ by construction, we have the inequality

$$F(x_n) < F(z) + \sqrt{\epsilon} d(x_n, z)$$

Our next step is to fix a unit tangent vector $v \in T_{x_n}M$ and let $\gamma : (-\alpha, \alpha) \to M$ be the unit speed geodesic with $\gamma(0) = x_n$ and $\frac{d}{dt}|_0 \gamma(t) = v$. We can take α to be small enough that γ realizes the distance between $\gamma(0)$ and all other $\gamma(t)$. From this assumption, it follows that $\gamma(t) \neq \gamma(0)$ for every nonzero $t \in (-\alpha, \alpha)$. Now rewrite our prior equation in terms of γ:

$$F(\gamma(0)) < F(\gamma(t)) + \sqrt{\epsilon_n} d(x_n, \gamma(t))$$

which we rearrange as

$$F(\gamma(0)) - F(\gamma(t)) < \sqrt{\epsilon_n} d(\gamma(0), \gamma(t)) = \sqrt{\epsilon_n}|t|$$

where the final equality is a result of our assumptions on γ. Since F is continuously differentiable,

$$F(\gamma(0)) - F(\gamma(t)) = \int_0^t dF(\gamma'(t))dt = \int_0^t \langle \nabla F, \gamma'(t) \rangle_g dt$$

So for all $t \in (-\alpha, \alpha)$,

$$\sqrt{\epsilon_n}(-t) < \int_0^t \langle \nabla F, \gamma'(t) \rangle_g dt < \sqrt{\epsilon_n} t$$

Hence, by differentiating at $t = 0$, we get $|\langle \nabla F, v\rangle_g| \leq \sqrt{\epsilon_n}$. Since this is true for any unit tangent vector at x_n, we conclude that

$$\|\nabla F(x_n)\| \leq \sqrt{\epsilon_n} \tag{4.4}$$

As $n \to \infty$, ϵ_n and δ_n both converge to 0. So all three conditions are clearly satisfied as a result of equations 4.3 and 4.4. $\square$

4.1.2 Proof of Gradient Estimates

We will introduce the following notation: $A \lesssim B$ indicates there is an absolute constant c such that $A \leq c \cdot B$. Now we are ready to define the key conditions functions under our consideration must satisfy:

Definition 4.5. (C1) $f : M \to [0, \infty) \in C^\infty(M, \mathbb{R})$ has only non-degenerate critical points

(C2) $f \lesssim \|\nabla f\|_g$

(C3) $f \lesssim \|\nabla f\|_g^2$

Remark 4.6. (C1) can be restated as f must be a non-negative Morse function. Condition (C2) is stronger than (C3) unless f is small. We will require both in order to derive stronger results about minima.

4.2 Uniqueness of Minima

The first key result we need about functions satisfying (C1)-(C3) is the uniqueness and existence of a minimum.

Theorem 4.7. *Let $f \in C^\infty(M, \mathbb{R})$ satisfy (C1) and (C2). Then:*

(i) A critical point of f must be a local minima, and can occur only when $f(x) = 0$.

(ii) f has at most one critical point, and $\inf_M f = 0$.

(iii) If f also sastisfies (C3), then f achieves the infimum, so it has exactly one critical point.

Proof. Condition (C2) tells us that if $df = 0$ at some $x \in M$, i.e. if f has a critical point, then $f(x) \lesssim 0$, so $f = 0$. Since $f \geq 0$, this means that x is a minimum.

Given that f satisfies (C1), we may apply proposition 4.4, which yields a sequence $\{x_n\}$ in M such that $f(x_n) \to \inf_M f$ and $\|\nabla f(x_n)\|_g \to 0$ as $n \to \infty$. Condition (C2) guarantees that $f(x_n) \lesssim \|\nabla f(x_n)\|_g$. Since $\|\nabla f(x_n)\|_g \to 0$, $f(x_n) \to 0$, hence $\inf_M f = 0$. Now we need to establish that f has at most one critical point.

Suppose that f has two critical points. Since (i) has been shown, they must necessarily both be minima. Call them x_0 and x_1. By (C1), the critical points are non-degenerate, so $d(x_0, x_1) > 0$. Since $f(x_0) = 0 = f(x_1)$, there must exist some $r, \alpha > 0$ such that $f(x) \geq \alpha$ for all $x \in \partial B(x_0, r)$ and $r < d(x_0, x_1)$. Then we can apply theorem 4.2, there exists a (PS)-sequence y_n such that $f(y_n) \to c \geq \alpha$. However, since this is a (PS)-sequence, $\|df(y_n)\|_g \to 0$ and because $\|\nabla f\|_g = \|df\|_g$, $\|\nabla f(y_n)\|_g \to 0$. However, (C2) implies that $f(y_n) \to 0$ as well, which is a contradiction since $0 < \alpha < \lim_{n\to\infty} f(y_n) = 0$. Thus, f has at most one critical point. So (ii) has been proved.

It remains to show that f achieves its minimum whenever it satisfies (C3). Our method will be to demonstrate that the gradient flow must converge to a minimum and it will do so in a finite time. We fix an $x_0 \in M$, and set $\gamma : [0, t_0) \to M$ to be the gradient flow: $\gamma(0) = x_0$ and $\gamma'(t) = -\nabla f(\gamma(t))$. Here, t_0 is taken to be the maximum time for which the flow exists.

We also define an intermediary function, g as $g(t) := f(\gamma(t))$, in order to simplify our notation. The derivative of g is $g'(t) = -\|\nabla f(\gamma(t))\|_g^2$. By the inequality (C3), we know that there exists some $c > 0$ such that $-cg'(t) = -c\|\nabla f(\gamma(t))\|_g^2 \geq f(\gamma(t)) = g(t)$. Setting $\epsilon = 1/c$, we rewrite this as $\epsilon g(t) \leq -g'(t)$. One more rearrangement yields $g'(t)/g(t) < -\epsilon$. For any interval $[a, b] \subset [0, t_0)$, we can simply integrate:

$$\int_a^b \frac{g'(t)}{g(t)} dt < -\epsilon(b - a)$$

The integral on the left is a logarithmic integral, so $\log(g(b)) - \log(g(a)) = \log(g(b)/g(a)) \leq \epsilon(b-a)$, and we can exponentiate to arrive at the inequality $g(b) \leq g(a)e^{-\epsilon(b-a)}$. In particular, we have

$$g(a) \leq e^{-\epsilon a} g(0) \tag{4.5}$$

by taking $[a, b]$ to be $[0, a]$ for any $a < t_0$.

For an interval $[a, b]$, the distance that γ travels is given by $\int_a^b \|\gamma'(t)\|_g dt$. We use Cauchy-Schwartz to estimate this value:

$$\begin{aligned}
\left(\int_a^b \|\gamma'(t)\|_g dt \right)^2 &= \left[\int_a^b \|\gamma'(t)\|_g^2 \right] \cdot (b - a) \\
&= \left[\int_a^b |g'(t)| dt \right] \cdot (b - a) \\
&= (g(a) - g(b))(b - a) \leq g(a)(b - a)
\end{aligned}$$

Notice that as $g'(t) < 0$, $|g'(t)| = -g'(t)$, and $\int_a^b -g'(t) = -(g(b) - g(a))$. If $b - a = 1$, then we can bound the square of the length of γ on that interval by $g(a)$, and by 4.5, the squared length is then bounded by $e^{-\epsilon a} g(0)$. So we have:

$$\int_a^{a+1} \|\gamma'(t)\|_g dt \leq e^{-\epsilon a/2} g(0)^{1/2} \tag{4.6}$$

Now for any interval $[x', y'] \subset [0, t_0)$, we can replace x with $\lfloor x \rfloor$ and y with $\lceil y \rceil$ (as long as $\lceil y \rceil < t_0$, but this does not affect the esimate), and divide into N consecutive intervals of length 1 as

$$[x, x+1], [x+1, x+2], \ldots, [x+n, x+n+1], \ldots, [x+N-1, x+N]$$

where $N = y - x$. Now we apply the estimate from the previous paragraph, equation 4.6:

$$\begin{aligned} d(\gamma(y), \gamma(x)) &\leq \sum_{n=0}^{N} d(\gamma(x+n), \gamma(x+n+1)) \\ &= \sum_{n=0}^{N} \int_{x+n}^{x+n+1} \|\gamma'(t)\|_g dt \\ &\leq \sum_{n=0}^{N} e^{-\epsilon(x+n)/2} g(0)^{1/2} \\ &= g(0)^{1/2} \sum_{n=0}^{N} e^{-\epsilon(x+n)/2} \\ &\leq g(0)^{1/2} \sum_{n=0}^{\infty} e^{-\epsilon(x+n)/2} \\ &= g(0)^{1/2} \frac{e^{(-\epsilon x/2)}}{1 - e^{-\epsilon/2}} \\ &\leq g(0)^{1/2} \frac{1}{1 - e^{-\epsilon/2}} \end{aligned} \tag{4.7}$$

In particular, γ will stay in a compact set, independent of x and y. Thus, the gradient flow exists for all positive time t. Thus, the limit point of $\gamma(t)$, which must be critical, exists. This completes the proof of (iii). $\square$

Now that we have this result, we are ready to prove the key growth estimate that functions satisfying conditions (C1)-(C3) satisfy. This inequality is one of the core components in the proof of the Anosov condition that will be the focus of Chapters 5 and 6.

4.3 Exponential Growth Estimate

Proposition 4.8. *Let $f \in C^\infty(M, \mathbb{R})$ satisfy (C1)-(C3) and x_0 be the unique minimum of f. Then there exists uniform constants, $\epsilon, C_1, C_2 > 0$ such that*

$$f(x) \geq \frac{1}{C_1} e^{\epsilon \cdot d(x_0, x)} - C_2$$

Proof. Our first step will be to prove the following inequality

$$d(x_0, x) \leq c_1 + \frac{1}{\epsilon} \ln(1 + f(x))$$

Let $x' \in M$ be a point where $f(x') \leq 2$. As in the proof of Theorem 4.7, we take the gradient flow $\gamma : [0, \infty) \to M$, where $\gamma'(t) = -\nabla f(\gamma(t))$ and then using the estimate in Equation 4.7, we can estimate $d(x_0, x')$:

$$d(x_0, x') \leq \lim_{t \to \infty} d(\gamma(t), \gamma(0)) \leq c_3$$

where c_3 is a universal constant from Equation 4.7. So for any x' satisfying $f(x') \leq 2$, $d(x_0, x')$ is bounded by a universal constant c_3.

Now, let γ begin at a point $x = \gamma(0)$, satisfying $f(x) > 2$. Let $k = \lfloor(\log_2(f(x)))\rfloor$, and define a sequence $0 = t_0 < t_1 < t_2 < \cdots < t_k$ where t_j satisfies $f(\gamma(t_j) = f(x)/2^j$. In particular, for all j, $f(t_j) > 1$. We know such a sequence exists because the gradient flow will get arbitrarily close to the minimum 0.

We set $x_{j+1} = \gamma(t_j)$ for $0 \leq j \leq k$. We add 1 to the index to avoid overloading the notation of x_0, since that is already set to be the global minimum. Our next step is to show that we have the inequality:

$$d(x_j, x_{j+1}) \leq c_4$$

for some constant c_4 independent of j. We shall show this for the case of $0 = t_0$ and t_1, and then all the other cases follow by reparametrizing the path so that t_{j-1} is sent to 0.

We set $g(t) = f(\gamma(t))$ as in the proof of Theorem 4.7, and once again, this gives $g'(t) = -\|\nabla f(\gamma(t))\|_g^2$. We will estimate

$$\int_0^t |g'(t)|^{1/2} dt$$

and we will use the inequality from (C3) to do so. From (C3), we get the inequality $-g'(t) \geq \epsilon g(t)^2$ for a global constant $\epsilon > 0$. A simple computation of $\frac{d}{dt} g(t)^{-1}$ combined with the preceeding inequality gives

$$\frac{d}{dt}\frac{1}{g(t)} \geq \epsilon$$

We can now integrate from 0 to t_1:

$$\epsilon(t_1 - 0) \leq \int_0^{t_1} \frac{d}{dt}\frac{1}{g(t)}dt = \frac{1}{g(t_1)} - \frac{1}{g(0)} = \frac{2}{g(0)} - \frac{1}{g(0)} = \frac{1}{g(0)}$$

This equality tells us that $t_1 \leq \frac{1}{g(0)\epsilon}$. Since $g(0) \geq 2$, t_1 is uniformly bounded. This same argument works for the distance between any t_{j-1} and t_j, since $g(t_{j-1}) > 2$ for all $j \leq k$, we shall denote it as c_4'. Our next step is to apply Cauchy-Schwartz again:

$$\begin{aligned}
\left(\int_0^{t_1} |g'(t)|^{1/2}dt\right)^2 &= \left(\int_0^{t_1} \left(|g'(t)|(t+1)\right)^{1/2} \frac{1}{(t+1)^{1/2}}dt\right)^2 \\
&\leq \left(\int_0^{t_1} |g'(t)|(t+1)dt\right)\left(\int_0^{t_1} \frac{1}{t+1}dt\right) \\
&\leq (t_1+1)\left(\int_0^{t_1} |g'(t)|dt\right)\log(1+t_1) \\
&= (t_1+1)\frac{g(0)}{2} \cdot \log(1+t_1)
\end{aligned}$$

Now, since $\log(1+x) \leq x$ for $x \geq 0$ and we already showed that $t_1 < c_4'$, we can get a uniform bound:

$$\begin{aligned}
\left(\int_0^{t_1} |g'(t)|^{1/2}dt\right)^2 &\leq (t_1+1)\frac{g(0)}{2} \cdot \log(1+t_1) \\
&\leq \left(c_4'+1\right) \cdot \frac{g(0)}{2} \cdot \frac{1}{g(0)\epsilon} \\
&= \frac{1}{2\epsilon}\left(c_4'+1\right) < c_4^2
\end{aligned} \tag{4.8}$$

for some constant c_4. Since this only depends on $f(\gamma(t_j)) > 2$ and $f(\gamma(t_{j-1})) = f(\gamma(g_j))/2$, we now have the desired bound for all $j \leq k$:

$$d(x_j, x_{j+1}) \leq \int_{t_{j-1}}^{t_j} \|\gamma'(t)\|_g dt = \int_{t_{j-1}}^{t_j} \|\nabla f(\gamma(t))\|_g dt = \int_{t_{j-1}}^{t_j} |g'(t)|^{1/2}dt < c_4$$

Now we can bound $x = x_0$ and x_k, where, $f(x_k) \in [1, 2)$ and $f(x_0) > 2$:

$$d(x_1, x_{k+1}) \leq \sum_{j=1}^{k} d(x_j, x_{j+1}) \leq \sum_{j=1}^{k} c_4 = c_4 k \leq c_4 \log_2(f(x_0))$$

And now we can add the distance between x_{k+1} and x_0, since $f(x_{k+1}) < 2$, to achieve the desired inequality. Note that we include a $+1$ term inside the logarithm to ensure it holds for $x < 1$.

$$d(x_0, x_1) \leq d(x_0, x_{k+1}) + d(x_{k+1}, x_1) \leq c_3 + c_4 \log_2(1 + f(x_1))$$

So for any $x \in M$, we now have the inequality:

$$d(x_0, x) \leq c_3 + c_4 \log_2(1 + f(x))$$

Up to a rescaling and relabeling of constants, our initial goal of showing

$$d(x_0, x) \leq c_1 + \frac{1}{\epsilon} \ln(1 + f(x))$$

is done. Now we rearrange the equation into the form $\epsilon \cdot d(x_0, x) - \epsilon c_1 \leq \ln(1 + f(x))$. Then we exponentiate both sides:

$$1 + f(x) \geq e^{-\epsilon c_1} e^{\epsilon d(x_0, x)}$$

and after rearranging once more, we have:

$$f(x) \geq \frac{1}{C_1} e^{\epsilon \cdot d(x_0, x)} - C_2$$

which completes the proof. $\square$

Remark 4.9. This proof also holds for an isotropic v which achieves it's minimum, so if v is isotropic and known to achieve a minimum, then it satisfies the same inequality.

Chapter 5

SU(n+1,1) Higgs Bundles

The focus of this chapter will be $(SU)[n+1,1]$-Higgs bundles. We will provide a new proof that the representations are convex cocompact, using the fact that Anosov representations in rank 1 are convex cocompact. Additionally, we will use the data of Higgs bundle to construct surfaces with $\mathbb{CH}^n$ and $\partial\mathbb{CH}^n$ structures. In particular, $\partial\mathbb{CH}^n$-structures correspond to the Guichard-Weinhard structures associated to the representation as an Anosov representation.

5.1 Conventions on Riemann Surfaces

In this section we will establish the conventions and structures associated to Riemann surfaces from this point forwards. These will be applied to the Riemann surfaces in this chapter, as well as in the later chapters.

We will always assume X to be a compact Riemann surface of genus $g \geq 2$. We will fix a basepoint x_0, which yields a universal cover $\widetilde{X}$, along with a fundamental group $\pi_1(X, x_0)$ but we will supress the x_0 in the notation and generally write $\pi_1(X)$.

The Riemann surface X has genus $g \geq 2$, so $\widetilde{X}$ is isomorphic to the open unit disk $D \subset \mathbb{C}$. The disk is a hyperbolic Riemann surface under the Poincaré disk model and so has a natural hyperbolic metric. In fact we then fix an isometry between the $\widetilde{X}$ with a hyperbolic metric and $\mathbb{H}^2$, the hyperbolic plane. As $\pi_1(X)$ acts discretely on $\widetilde{X}$ via deck transforms, it is a well known fact that $X \cong \widetilde{X}/\pi_1(X)$. The identification of $\widetilde{X}$ with $\mathbb{H}^2$ also yields a an isomorphism between $\pi_1(X)$ and a discrete subgroup of the isometry group of $\mathbb{H}^2$, the projective linear group $\mathsf{PSL}(2, \mathbb{R})$. So we have $\pi_1(X) \cong \Gamma \subset \mathsf{PSL}(2, \mathbb{R})$. Such a group is a special case of a *Fuchsian group* (it is torsion free, whereas a general Fuchsian group, a discrete subgroup of $\mathsf{PSL}(2, \mathbb{R})$, can have torsion). We call this isomorphism $\rho_F : \pi_1(X) \to \Gamma$ a *Fuchsian representation*, where the F subscript stands for Fuchsian. Hence,

$$X \cong \mathbb{H}^2/\Gamma$$

and as Γ is a subgroup of isometries, X inherits a natural hyperbolic metric. Thus, X and $\widetilde{X}$ have complete hyperbolic metrics which we shall denote with $\|\cdot\|$.

5.2 SU(n+1,1) Lie Theory Basics

Our first step will be to establish that the holonomy representation of an $\mathsf{SU}(n+1,1)$-Higgs bundle $(\mathcal{E}, \varphi)$ over a Riemann surface X with genus $g \geq 2$ is Anosov, and hence convex cocompact. Before we begin the proof, we will establish the conventions we will use.

We will define H, a unitary matrix with the form:

$$H = \begin{bmatrix} I_{n+1} & 0 \\ 0 & -1 \end{bmatrix}$$

H defines a signature $(n+1, 1)$ Hermitian form on $\mathbb{C}^{n+1,1}$ by $w^* H v$ for $w, v \in \mathbb{C}^{n+1,1}$ where w^* signifies the conjugate transpose of w. We define $\mathsf{SU}(H)$ to be the matrices $A \in \mathsf{GL}(n+2, \mathbb{C})$ such that $A^* H A = H$. As H has signature $(n+1, 1)$, we will simplify notation to $\mathsf{SU}(H) = \mathsf{SU}(n+1, 1)$.

The Lie algebra of $\mathsf{SU}(n+1, 1)$ is given by

$$\mathfrak{su}(n+1, 1) = \{X \in \mathfrak{sl}(n+2, \mathbb{C}) \mid X^* H + H X = 0\}$$

We can write an element X in the Lie algebra in block form as

$$X = \begin{bmatrix} a & b \\ b^* & d \end{bmatrix}$$

where a is an $(n+1) \times (n+1)$ skew Hermitian matrix, and d is a 1×1 skew Hermitian matrix, i.e. d is a purely imaginary number. We also require the condition that $\mathrm{Tr}\, a + \mathrm{Tr}\, d = 0$ in order to lie inside $\mathfrak{sl}(n+2, \mathbb{C})$. We define the Cartan involution θ to by $\theta : X \mapsto -X^*$. Then:

$$\theta(X) = \begin{bmatrix} -a^* & -b \\ -b^* & -d^* \end{bmatrix}$$

From which, we can see the fixed points are matrices with $b = 0$, and matrices with $a = 0$ and $d = 0$ are multiplied by -1. Hence the ± 1-eigenspaces of the Cartan decomposition are given by:

$$\mathfrak{l} = \left\{ \begin{bmatrix} a & b \\ b^* & d \end{bmatrix} \in \mathfrak{su}(n+1,1) \;\middle|\; b = 0 \right\} = \left\{ \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \in \mathfrak{su}(n+1,1) \right\}$$

$$\mathfrak{p} = \left\{ \begin{bmatrix} a & b \\ b^* & d \end{bmatrix} \in \mathfrak{su}(n+1,1) \;\middle|\; a = 0, d = 0 \right\} = \left\{ \begin{bmatrix} 0 & b \\ b^* & 0 \end{bmatrix} \in \mathfrak{su}(n+1,1) \right\}$$

So we have our Cartan decomposition $\mathfrak{su}(n+1,1) = \mathfrak{l} \oplus \mathfrak{p}$. The maximal compact subgroup of $\mathsf{SU}(n+1,1)$ is then given by exponentiating $\mathfrak{l}$, which we can now see is a copy of $\mathsf{S}(\mathsf{U}(n+1) \times \mathsf{U}(1))$.

We are working with a finite dimensional real semisimple Lie algebra over $\mathbb{C}$, so the Cartan subalgebra is equivalent to a maximal abelian subalgebra. Then one way we can define $\mathfrak{a}$, the Cartan subalgebra, is by matrices in $\mathfrak{p}$ where the block b has the form:

$$b = \begin{bmatrix} \beta \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

where β is a non-zero real number. Let $\alpha \in {}^*a$, be defined by:

$$\alpha\left(\begin{bmatrix} 0 & b \\ b^* & 0 \end{bmatrix} \right) = \beta$$

We will prefer to work with elements of $\mathfrak{a}$ that are diagonal, so we will perform a change of basis. Define the matrices:

$$V = \left[\begin{array}{c|ccc|c} 1 & 0 & \cdots & 0 & 1 \\ \hline 0 & & & & 0 \\ \vdots & & \mathrm{Id}_n & & \vdots \\ 0 & & & & 0 \\ \hline 1 & 0 & \cdots & 0 & -1 \end{array}\right], \quad V^{-1} = \left[\begin{array}{c|ccc|c} \frac{1}{2} & 0 & \cdots & 0 & \frac{1}{2} \\ \hline 0 & & & & 0 \\ \vdots & & \mathrm{Id}_n & & \vdots \\ 0 & & & & 0 \\ \hline \frac{1}{2} & 0 & \cdots & 0 & -\frac{1}{2} \end{array}\right]$$

Under this change of basis, we compute the following, where $X \in \mathfrak{a}$:

$$V^{-1}HV = \begin{bmatrix} 0 & 0 & 1 \\ 0 & \mathrm{Id}_n & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

and $V^{-1}XV = \mathrm{diag}(-\beta, 0, \ldots, 0, \beta)$. From here on, we will use H to mean the above matrix, and assume that we are working in this new basis. In particular, our basis is given by

$$e_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \vdots \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad e_{n+2} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \vdots \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

and $e_2, \dots, e_{n+1}$ are the standard basis vectors. This forms an orthonormal basis with respect to the Hermitian form H. The vectors $e_1, \dots, e_{n+1}$ are positive with respect to the Hermitian form, and e_{n+2} is negative. In terms of this basis, we can write down the weight spaces for the roots $\alpha, -\alpha \in a^*$ for the standard representation. We have:

$$W_\alpha = \operatorname{span}\left(\frac{1}{\sqrt{2}}(e_1 - e_{n+2})\right), \qquad W_{-\alpha} = \operatorname{span}\left(\frac{1}{\sqrt{2}}(e_1 + e_{n+2})\right)$$

5.3 SU(n+1,1)-Higgs Bundles and the Key Function

Now we have established the Lie theoretic groundwork. We will next consider an $\mathsf{SU}(n+1,1)$-Higgs bundle (E, φ). Recall that such a bundle over a Riemann surface X of genus $g \geq 2$ is given by the data of a tuple (V, W, β, γ) such that: V and W are respectively rank p and q holomorphic vector bundles satisfying $\deg V + \deg W = 0$ and the data of the Higgs field is given by $\beta \in H^0(X, \hom(W, V) \otimes K)$ and $\gamma \in H^0(X, \hom(V, W) \otimes K)$.

Our focus will be on polystable $\mathsf{SU}(n+1,1)$-Higgs bundles that are parametrized by a tuple $(\mathcal{V}_1, \mathcal{V}_2, \beta)$, where $\mathcal{V}_1$ is a rank n holomorphic vector bundle, $\mathcal{V}_2$ is a holomorphic line bundle and $\beta \in H_0(X, \hom(\mathcal{V}_1, \mathcal{V}_2) \otimes K)$. This yields a Higgs bundle of the form:

$$\mathcal{V}_1 \xrightarrow{\beta_1} \mathcal{V}_2 \xrightarrow{1} \mathcal{V}_3$$

where $\mathcal{V}_3 = \mathcal{V}_2 K^{-1}$ is a line bundle satisfying $\deg \mathcal{V}_1 + \deg \mathcal{V}_2 + \deg \mathcal{V}_3 = 0$, and $\mathcal{V}_1 \oplus \mathcal{V}_3 = V$ and $\mathcal{V}_2 = W$. The map γ decomposes into $\gamma_1 : \mathcal{V}_2 \to \mathcal{V}_1 \otimes K$ and $\gamma_3 : \mathcal{V}_2 \to \mathcal{V}_3 \otimes K$. We will assume that $\gamma_1 = 0$ and γ_3 is an isomorphism and so will be denoted by 1. We will also assume that this decomposition is orthogonal with respect to the harmonic metric.

Definition 5.1. A polystable $\mathsf{SU}(n+1,1)$-Higgs bundle which decomposes as above satisfying $\gamma_1 = 0$ and that γ_3 is an isomorphism and hence non-vanishing will be said to satisfy the *non-vanishing assumption*. We will denote γ_3 sometimes as 1 due to this fact.

This assumption will be vital and we will always assume to be working with Higgs bundles that satisfy this condition.

There exists a flat connection on the Higgs bundle coming from the non-abelian Hodge correspondence, ∇, which decomposes as:

$$\nabla = \nabla^{Ch} + \varphi + \varphi^*$$

where φ is the Higgs field, and decomposes as β_1 and γ_3. A key calculation we compute is that for a flat section $v \in \mathcal{V}_1 \oplus \mathcal{V}_2 \oplus \mathcal{V}_3$, if we apply the flat connection to v we get 0. In particular, if we look at what happens to the component v_3 in $\mathcal{V}_3$, we have:

$$0 = (\nabla v)_3 = \nabla^{Ch} v_3 + \varphi(v_2)$$

since φ^* does not map anything into the $\mathcal{V}_3$. The key takeaway is that we can understand the Chern connection in terms of the Higgs field:

$$\nabla^{Ch} v_3 = -\varphi(v_3) \tag{5.1}$$

In order to work with the Higgs bundle, we will fix a basepoint $x_0 \in X$, which we will identify with its lift in the $\widetilde{X}$, and then we will pullback E over the universal cover:

$$\begin{array}{ccc} \pi^*E & \longrightarrow & E \\ \downarrow & & \downarrow \\ \widetilde{X} & \xrightarrow{\pi} & X \end{array}$$

Since we will be clear when we are working on the pullback over the universal cover or over the Riemann surface X, we will simplify notation and refer to both bundles with the same symbols. By definition of the holonomy representation, for $\gamma \in \pi_1(X)$, a point $(x, v) \in \pi^*E$ the action of γ on x takes corresponds to $\rho(\gamma)$ acting on v. Since $\rho(\gamma) \in \mathsf{SU}(n+1, 1)$, $\|\rho(\gamma)v\|_{h(\gamma x)} = \|v\|_{h(x)}$.

For any flat section v, we can identify it, and will again overload notation, with the vector $v \in E(x_0)$, where $E(x_0)$ is the fiber over x_0. This is done using the harmonic metric and parallel transport.

Now we come to a very important construction. Recall from Theorem 3.12 that the harmonic metric that comes from the poly-stability of E is compatible with the splitting of E into $(\mathcal{V}_1 \oplus \mathcal{V}_3) \oplus \mathcal{V}_2$, so the harmonic metric (which is Hermitian) H splits as $h = h_{\mathcal{V}_1 \oplus \mathcal{V}_3} \oplus h_{\mathcal{V}_2}$ and the indefinite hermitian form coming from the $\mathsf{SU}(n+1, 1)$ structure is $h_I = h_{\mathcal{V}_1 \oplus \mathcal{V}_3} \oplus -h_{\mathcal{V}_2}$. We will denote the norm with respect to these forms as $\|v\|_h = h(v, v)$ and $\|v\|_{h_I} = h_I(v, v)$.

Let $v \in E(x_0)$ be a non-zero negative or isotropic vector with respect to the indefinite Hermitian metric. We decompose v according the orthogonal decomposition of the bundle, into components $v = v_1 + v_2 + v_3$. A key fact to notice is that since the decomposition is orthogonal with respect to both the definite and indefinite Hermitian forms.

$$h_I(v,v)_{x_0} = \|v_1\|_{h(x_0)} - \|v_2\|_{h(x_0)} + \|v_3\|_{h(x_0)} \leq 0$$

and since $\|v_1\|_{h(x_0)}, \|v_2\|_{h(x_0)} \geq 0$ and h is non-degenerate, it must be the case that $v_2 \neq 0$, since $\|v_2\|_h > \|v_1\|_h + \|v_2\|_h$ and v is non-zero. Furthermore, if v is negative, we can assume it has been scaled such that $\|v\|_{h_I} = -1$. As we vary the basepoint away from x_0 to another point x, the decomposition of $\mathcal{V}_1(x_0) \oplus \mathcal{V}_2(x_0) \oplus \mathcal{V}_3(x_0)$ changes to $\mathcal{V}_1(x) \oplus \mathcal{V}_2(x) \oplus \mathcal{V}_3(x)$. Now the norm remains fixed, i.e. $\|v\|_{h_I(x_0)} = \|v\|_{h_I(x)}$, but the decomposition into components changes. Let $v_3(x)$ represent the v_3 component of v relative to the decomposition at x. We can now introduce a very important function.

Definition 5.2. Let $v \in E(x_0)$ be a non-zero negative or isotropic vector. Then we define $f_v : \widetilde{X} \to [0, \infty)$ by $f_v(x) = h(v_3(x), v_3(x))_x = \|v_3(x)\|_{h(x)}$ to be the norm of $v_3(x)$ with respect to the metric at x.

Under the non-vanishing assumption, $\gamma_3 : \mathcal{V}_2 \to \mathcal{V}_3 \otimes K$ is nowhere 0. Further, $v_2(x)$ cannot vanish either. Thus there exists a vector field V_v such that

$$\gamma_3(V_v)v_2 = -v_3$$

Indeed, $\gamma_3 v_2$ yields an element of $\mathcal{V}_3 \otimes K$ which is nowhere 0, and as $\mathcal{V}_3$ is a line bundle, there exists some vector field on X such that when it is paired with the contangent vector field in K, always scales $\gamma_3 v_2$ so that it equals $-v_3$. We will call V_v the *directing vector field of f_v*. Our next goal is to establish that f_v satisfies the conditions of definition 4.5.

Lemma 5.3. *For a nonzero nonpositive vector $v \in E(x_0)$, the function f_v satisfies conditions (C1) and (C2) of Definition 4.5. If v is a negative vector, then (C3) is also satisfied.*

Proof. First we will prove that f_v only has non-degenerate critical points. We compute the Hessian $Hf_v(X, X)$ for X, Y vector fields on $\widetilde{X}$. First, as the Chern connection is unitary, we have:

$$dh(v_3, v_3) = h(\nabla^{Ch} v_3, v_3) + h(v_3, \nabla^{Ch} v_3)$$

So $df_v = h(\nabla^{Ch} v_3, v_3) + h(v_3, \nabla^{Ch} v_3)$. We can apply this twice in the in the computation of the Hessian.

$$\begin{aligned}
Hf_v(X, Y) &= d(df_v(X))(Y) \\
&= d(h(\nabla_X^{Ch} v_3, v_3) + h(v_3, \nabla_X^{Ch} v_3))(Y) \\
&= h(\nabla_Y^{Ch} \nabla_X^{Ch} v_3, v_3) + h(\nabla_X^{Ch} v_3, \nabla_Y^{Ch} v_3) + h(\nabla_Y^{Ch} v_3, \nabla_X^{Ch} v_3) + h(v_3, \nabla_Y^{Ch} \nabla_X^{Ch} v_3)
\end{aligned}$$

At a critical point of f_v, i.e. a point x where $v_3(x) = 0$, so the terms $h(\nabla_Y^{Ch} \nabla_X^{Ch} v_3(x), v_3(x)) = 0$ and $h(v_3(x), \nabla_Y^{Ch} \nabla_X^{Ch} v_3(x)) = 0$. Thus, $Hf_v(X, Y) = h(\nabla_X^{Ch} v_3, \nabla_Y^{Ch} v_3) + h(\nabla_Y^{Ch} v_3, \nabla_X^{Ch} v_3)$. Now to verify that the Hessian is positive definite, we have:

$$Hf_v(X, X) = h(\nabla_X^{Ch} v_3, \nabla_X^{Ch} v_3) + h(\nabla_X^{Ch} v_3, \nabla_X^{Ch} v_3) = 2\|\nabla_X^{Ch} v_3\|^2 = 2\|\gamma_3(X) v_2\|^2 > 0$$

for any vector field X which is non-zero at the critical point, since γ_3 and v_2 are both non-vanishing. Thus, since the Hessian is positive definite at a critical point, f_v has no degenerate critical points and (C1) is satisfied.

Next we will verify (C2) and (C3). First, we recall the calculation of df_v and combine it with the fact that $\nabla^{Ch} v_3 = -\gamma_3 v_2$:

$$df_v = dh(v_3, v_3) = h(\nabla^{Ch} v_3, v_3) + h(v_3, \nabla^{Ch} v_3) = h(-\gamma_3 v_2, v_3) + h(v_3, -\gamma_3 v_2) \tag{5.2}$$

We recall that the directing vector field was defined with the property that $-\gamma_3(V_v)v_2 = v_3$. Thus

$$df_v(V_v) = h(v_3, v_3) + h(v_3, v_3) = 2f_v$$

We also observe from Equation 5.2, the terms $h(-\gamma_3 v_2, v_3)$ and $h(v_3, -\gamma_3 v_2)$ are conjugate to each other. So we can compute

$$\begin{aligned} df_v(\sqrt{-1}V_v) &= h(-\gamma_3(\sqrt{-1}V_v)v_2, v_3) + h(v_3, -\gamma_3(\sqrt{-1}V_v)v_2) \\ &= -\sqrt{-1}h(-\gamma_3(V_v)v_2, v_3) + \sqrt{-1}h(v_3, -\gamma_3(V_v)v_2) \\ &= -\sqrt{-1}h(v_3, v_3) + \sqrt{-1}h(v_3, v_3) \\ &= 0 \end{aligned}$$

which shows that $\sqrt{-1}V$ is orthogonal to to ∇f_v. Since these are both vector fields on $\widetilde{X}$, which dimension 2, it means that since $\sqrt{-1}V$ and V are also orthogonal, V and ∇f_v are proportional to each other. Now, we had already shown that $df_v(V_v) = 2f$, and it is a standard geometry result that $df_v(\nabla f_v) = \|\nabla f_v\|^2$. As ∇f_v and V_v are proportional, either they are both 0 or $\nabla f_v = \frac{\|\nabla f_v\|}{\|V_v\|}V_v$. Thus, $df_v(\nabla f_v) = \frac{\|\nabla f_v\|}{\|V_v\|}df_v(V_v)$. We can combine all these equations now to get the relation:

$$\|\nabla f_v\| \cdot \|V_v\| = 2f$$

Now we need to bound $\|V_v\|$. Since X is compact, $\|\gamma_3\|$ is bounded. Since $|h(v_3, v_3)| \leq |h(v_2, v_2)|$ (else v would be positive), we know that $\|v_3\| \leq \|v_2\|$. Since V_v is defined to scale $-v_3/\gamma_3 v_2$, and γ is bounded while the norm of v_3 is less than or equal to the norm of v_2, $\|V_v\|$ must be bounded, hence $\|V_v\| \lesssim 1$. Thus, since $\|\nabla f_v\| \cdot \|V_v\| = 2f_v$, we can conclude that $f \lesssim \|\nabla f_v\|$, so condition (C2) is satisfied.

All that remains is to show condition (C3). Since $h(v, v) = -1$, $h(v_2, v_2) \leq -1$, so $\|v_2\| \geq 1$. Moreover, $\|v_3\| = \sqrt{h(v_3, v_3)} = f^{1/2}$. So once again recalling that $V_v = -v_3/\gamma_3 v_2$, we have that $\|V_v\| \lesssim f_v^{1/2}$. Once again we plug this into $\|\nabla f_v\| \cdot \|V_v\| = 2f_v$ and get $f_v^{1/2} \lesssim \|\nabla f_v\|$. This completes the proof. $\square$

5.4 Geometric Structures and Domains of Discontinuity

Our goal in this next section is to describe the geometric structures on bundles relate to the Riemann surface X that arise from the $\mathsf{SU}(n+1,1)$-Higgs bundles considered above, i.e. ones that decompose as above and satisfy the non-vanishing assumption. Fix the associated representation $\rho : \pi_1(X) \to \mathsf{SU}(n+1,1)$. The key idea is that we will construct a bundle over $\widetilde{X}$ using negative or isotropic lines that are perpendicular to $\mathcal{V}_3$ in the fiber above each point, and then construct a natural map to complex hyperbolic space, $\mathbb{CH}^n$, or the boundary of complex hyperbolic space, $\partial\mathbb{CH}^n$. We will prove that this map is a developing map, and thus induces a geometric structure on the bundle. Throughout this section, the properties of the function f_v, in particular the existence (if v is negative) and uniquness of its minima will play a fundamental role.

Our first task is to define the bundles on which we will construct geometric structures. As always, recall we fixed an $x_0 \in \widetilde{X}$. To start, for any $x \in \widetilde{X}$, the fiber above X in the vector bundle formulation of the Higgs bundle naturally has a $\mathbb{C}^{n+1,1}$ structure. Then, we define the following:

1.
$$S_x = \{v \in E(x_0) \mid \|v\|_{h_I(x_0)} = 0 \text{ and } v \perp \mathcal{V}_3(x)\}$$

   That is, we define a S_x to be the set of isotropic vectors in $E(x_0)$ that live entirely in $\mathcal{V}_1(x) \oplus \mathcal{V}_2(x)$ relative to the splitting over x. As $\mathcal{V}_1(x) \oplus \mathcal{V}_2(x)$ is isomorphic to $\mathbb{C}^{n,1}$, S_x is isomorphic to the space of isotropic vectors in $\mathbb{C}^{n,1}$.

2.
$$D_x = \{v \in E(x_0) \mid \|v\|_{h_I(x_0)} < 0 \text{ and } v \perp \mathcal{V}_3(x)\}$$

   This is the set of negative vectors in $E(x_0)$ entirely contained in the splitting $\mathcal{V}_1(x) \oplus \mathcal{V}_2(x)$ over x. Similar to the above case, D_x is isomorphic to the space of negative vectors in $\mathbb{C}^{n,1}$.

Recall that the projective model for complex hyperbolic n-space is defined as the negative lines, and the boundary is defined as isotropic lines in a copy of $\mathbb{C}^{n,1}$. That is, for each x, $\mathbb{P}S_x \cong \partial\mathbb{CH}^n$ and $\mathbb{P}D_x \cong \mathbb{CH}^n$.

Definition 5.4. Define a bundle $\widetilde{\mathcal{S}} \to \widetilde{X}$ whose fiber above $x \in \widetilde{X}$ is $\mathbb{P}S_x$. The total space of this bundle is naturally a subset of $\widetilde{\mathcal{S}} \subset \widetilde{X} \times \partial\mathbb{CH}^{n+1}$.

Definition 5.5. Define a bundle $\widetilde{\mathcal{D}} \to \widetilde{X}$ whose fiber over $x \in \widetilde{X}$ is $\mathbb{P}D_x$. The total space of this bundle is naturally a subset of $\widetilde{\mathcal{D}} \subset \widetilde{X} \times \mathbb{CH}^{n+1}$

These bundles naturally descend to bundles over X, as we see in the following proposition.

Proposition 5.6. *The action of $\pi_1(X)$ on $\widetilde{X} \times \partial\mathbb{CH}^{n+1}$ and $\widetilde{X} \times \mathbb{CH}^{n+1}$, where the action on the second factor is given by ρ, leaves $\widetilde{\mathcal{S}}$ and $\widetilde{\mathcal{D}}$, respectively, invariant. Hence there exists bundles:*

$$\mathcal{S} := \left(\widetilde{\mathcal{S}}/\pi_1(X)\right) \to \left(\widetilde{X}/\pi_1(X)\right) = X$$

$$\mathcal{D} := \left(\widetilde{\mathcal{D}}/\pi_1(X)\right) \to \left(\widetilde{X}/\pi_1(X)\right) = X$$

Proof. We shall prove the result for $\widetilde{X} \times \partial\mathbb{CH}^{n+1}$ and the bundle $\widetilde{\mathcal{S}}$. The proof for the $\widetilde{\mathcal{D}}$ is completely analagous.

Since $\partial\mathbb{CH}^{n+1}$ consists of isotropic lines in $\mathbb{C}^{n+1,1}$, there is a natural action of $\mathsf{SU}(n+1,1)$ on these lines by matrix multiplication. As an element of $\mathsf{SU}(n+1,1)$ preserves the signature $(n+1,1)$ indefinite hermitian norm on any vector, i.e. $h_I(Av, Av) = h_I(v, v)$ for $A \in \mathsf{SU}(n+1,1)$ and $v \in \mathbb{C}^{n+1,1}$, it clearly fixes the space of isotropic vectors, hence the space of isotropic lines.

Let (x, v) be an element of $\widetilde{\mathcal{S}}$, so $x \in \widetilde{X}$, and $v \in E(x)$ such that $\|v\|_{h_I(x)} = 0$, and $v \perp \mathcal{V}_3(x)$. As we just saw that $\mathsf{SU}(n+1,1)$ preserves norms, we know that for any $\gamma \in \pi_1(X)$, $\|\rho(\gamma)v\|_{h_I(\gamma\cdot x)} = 0$. Further, for any $w \in \mathcal{V}_3(x)$, we know $h_I(\rho(\gamma)v, \rho(\gamma)w)_{\gamma\cdot x} = h_I(v, w)_x = 0$ by 3.3. Thus, $\gamma \cdot (x, v) \in \widetilde{\mathcal{S}}$. Then the projection $\widetilde{\mathcal{S}} \to \widetilde{X}$ descends through the quotient to $\mathcal{S} \to X$ as stated in the proposition. $\square$

Now we have the following developing maps:

$$\mathrm{Dev}^S : \widetilde{\mathcal{S}} \longrightarrow \partial\mathbb{CH}^{n+1}$$

$$\mathrm{Dev}^D : \widetilde{\mathcal{D}} \longrightarrow \mathbb{CH}^{n+1}$$

both are defined by projection onto the second factor. By construction, both of these maps are $\pi_1(X)$-equivariant. We define $\Gamma = \rho(\pi_1(X))$. We have the following theorems.

Theorem 5.7. *There exists a unique open set $\Omega_\rho \subset \partial\mathbb{CH}^{n+1}$ such that Dev^S is a diffeomorphism between $\widetilde{\mathcal{S}}$ and Ω_ρ. Then Dev^S is a developing map and so induces a $\partial\mathbb{CH}^{n+1}$ structure on $\mathcal{S}$.*

Moreover, the action of Γ on Ω_ρ is properly discontinuous, and Ω_ρ/Γ is diffeomorphic to $\mathcal{S}$ and $\mathcal{S}$ is diffeomorphic to a S^{2n-1}-bundle.

There is a similar theorem for the bundle $\mathcal{D}$.

Theorem 5.8. *The map Dev^D is a diffeomorphism between $\widetilde{\mathcal{D}}$ and $\mathbb{CH}^{n+1}$ and so Dev^D is a developing map and induces a $\mathbb{CH}^{n+1}$ structure on $\mathcal{D}$.*

Moreover, the action of Γ on $\mathbb{CH}^{n+1}$ is properly discontinuous and $\mathbb{CH}^{n+1}/\Gamma$ is diffeomorphic to $\mathcal{D}$, which is diffeomorphic to a complex disk bundle B^{n+1} over X.

As both theorems have similar proofs, we shall start by proving Theorem 5.7 completely, and then in the proof of Theorem 5.8, we shall indicate where equivalent arguments hold and focus on the differences in the proofs.

Proof of Theorem 5.7. We will first show that the developing map is a smooth injection. Then we will verify the proper discontinuity of the action of Γ and exhibit the desired bundle diffeomorphisms. Lastly, we will show it is a diffeomorphism onto its image.

Smoothness of Dev^S is immediate as it is a projection map between manifolds of real dimension $2n+1$. Now we show that the map is injective. Suppose that $[v] \in \partial\mathbb{CH}^{n+1}$ is in the image of Dev^S. Then to that v, which must be an isotropic vector, we assign the function f_v as described above. Since $[v]$ is in the image of Dev^S, there is an $x \in \widetilde{X}$ where $f_v(x) = 0$, since that corresponds to v being perpendicular to $\mathcal{V}_3(x)$. From Theorem 4.7, we know that f_v can have at most a single minima, so that x must be the only x which contains v in it's fiber. Thus, $[v]$ is in the image of at most a single point. So Dev^S is injective.

Now that we know that Dev^S is injective, we identify the image of Dev^S, which we denote by Ω_ρ with $\widetilde{\mathcal{S}}$. The action of ρ on Ω_ρ is properly discontinuous. This follows because the identification is made by Dev^S, a map which is equivariant, and the action of $\pi_1(X)$ on $\widetilde{\mathcal{S}}$ is certainly properly discontinuous since $\pi_1(X)$ acts properly discontinuously on $\widetilde{X}$. Once we have shown that Dev^S is a diffeomorphism onto Ω_ρ, it immediately follows from the equivariance of Dev^S that Ω_ρ/Γ is diffeomorphic to the bundle $\mathcal{S}$.

Our next task is to identify $\mathcal{S}$ with an appropriate S^{2n-1} bundle. We will work on $\widetilde{\mathcal{S}}$, being careful so that the identification is done in an equivariant manner, in order to ensure all the results pass to $\mathcal{S}$. Over a point $x \in \widetilde{X}$, an element of the fiber is a projective class $[w, z] \in \mathbb{P}(\mathcal{V}_1(x) \oplus \mathcal{V}_2(x))$ where $w \in \mathbb{C}^n$ and $z \in \mathbb{C}^\times$, i.e. z is non-zero, satisfying $\|w\|_{h(x)} - \|z\|_{h(x)} = 0$. Without loss of generality, we can assume that $z = 1$, by multiplying z^{-1}. Then, any $[w, 1]$ is completely determined by $w \in \mathbb{C}^n$ where $\|w\|_{h(x)} = 1$. Hence the unit sphere of $\mathbb{C}^n$ parametrizes a fiber of $\widetilde{\mathcal{S}}$. As all the arguments are unchanged under an action of Γ, this provides the desired identification.

Finally, we will show that Dev^S is a local diffeomorphism. We will start by computing Dev^S in local coordinates. Fix a local coordinate z on $U \subset \widetilde{X}$, and also take nonvanishing local holomorphic sections $c_1' : U \to \mathcal{V}_1$ and $c_2' : U \to \mathcal{V}_2$. We then take the associated section of unit norm: $c_1(z) := c_1'(z)/\|c_1'(z)\|$ and $c_2(z) := c_2'(z)\|c_2'(z)\|$. To an element $w \in S^{2n-1}$, associate a rotation matrix A_w which will necessarily be unitary and thus lie in $\mathsf{SU}(n)$ such that A_w takes $(1, 0, \ldots, 0)$ to w. Then:

$$
\begin{aligned}
\widetilde{X} \times S^{2n-1} &\xrightarrow{\mathrm{Dev}^S} \partial\mathbb{C}\mathbb{H}^{n+1} \\
(z, w) &\longmapsto [A_w c_1(z) + c_2(z)]
\end{aligned}
\tag{5.3}
$$

Now we need to compute the tangent spaces of some $[v_0] = \mathrm{Dev}^S(z_0, w_0) \in \partial\mathbb{C}\mathbb{H}^{n+1}$ for some $z_0 \in U$. The first thing to notice is that for a vector space V and subspace W, $T_W \mathbb{P}V = \hom(W, W^\perp)$. So the tangent space $T_{[v_0]}\partial\mathbb{C}\mathbb{H}^{n+1}$ is the space of homomorphisms from $[v_0]$ to $[v_0]^\perp$ in the nonzero isotropic vectors in $\mathbb{P}\left(\mathcal{V}_1(z_0) \oplus \mathcal{V}_2(z_0) \oplus \mathcal{V}_3(z)\right)$. It is simple to see that $[v_0]^\perp$ must be $\mathbb{P}\left(\mathcal{V}_1'(z_0) \oplus \mathcal{V}_3(z_0)\right)$ where $\mathcal{V}_1'(z_0) = \mathcal{V}_1'(z_0) \cap [v_0]^\perp$ and there is no $\mathcal{V}_2$ term because being isotropic requires having trivial $\mathcal{V}_2$ component, so no part of $\mathcal{V}_2$ is perpendicular to $[v_0]$. It follows that:

$$
T_{[v_0]}\partial\mathbb{C}\mathbb{H}^{n+1} = \hom([v_0], \mathcal{V}_1'(z_0) \oplus \mathcal{V}_3(z_0))
$$

Now the derivative of the developing map can be written in local coordinates as:

$$
T_{z_0}\widetilde{X} \times T_{w_0}S^{2n-1} \xrightarrow{D_{(z_0,w_0)}\mathrm{Dev}^S} \hom([v_0], \mathcal{V}_1'(z_0) \oplus \mathcal{V}_3(z_0))
$$

Now we can decompose this into block form as:

$$\begin{bmatrix} D(T_{z_0}\widetilde{X} \to \hom([v_0], \mathcal{V}_3(z_0))) & 0 \\ D(T_{z_0}\widetilde{X} \to \hom([v_0], \mathcal{V}_1'(z_0))) & D(T_{w_0}S^{n-1} \to \hom([v_0], \mathcal{V}_1'(z_0))) \end{bmatrix}$$

where it is immediately apparent from Equation 5.3 that the derivative in terms of w cannot move in $\mathcal{V}_3(z_0)$ component of the tangent space. Now in order to check that Dev^S is a local diffeomorphism, we must verify that this derivative is an isomorphism. It suffices to show that the diagonal is non-degenerate. First, it is clear that differentiating in a direction of w will produce a non-trivial variation in the $\mathcal{V}_1'(z)$ direction. So all that remains is to verify that $D(T_{z_0}\widetilde{X} \to \hom([v_0], \mathcal{V}_3(z_0)))$ is nondegenerate. To do this, we will compute the differential using the flat connection ∇ applied to Equation 5.3. Recall that the $\nabla = \nabla^{Ch} + \Phi + \Phi^\dagger$. We only care about variation in the direction of $\mathcal{V}_3(z_0)$, so our computation will be modulo the $\mathcal{V}_1(z_0)$ and $\mathcal{V}_2(z_0)$ components.

$$\nabla(A_w c_1 + c_2) = A_w \nabla c_1 + \nabla c_2$$

and we now compute:

$$\nabla c_1 = \nabla^{Ch} c_1 + \beta_1 c_1 = 0 \mod \mathcal{V}_1(z_0) \oplus \mathcal{V}_2(z_0)$$

$$\nabla c_2 = \nabla^{Ch} c_1 + \gamma_3 c_2 + \beta_1^\dagger c_2 = \gamma_3 c_2 \mod \mathcal{V}_1(z_0) \oplus \mathcal{V}_2(z_0)$$

So $D(T_{z_0}\widetilde{X} \to \hom([v_0], \mathcal{V}_3(z_0)))$ is given by $\gamma_3 c_2$, which, by the nonvanishing assumption, is nondegenerate. Hence, Dev^S is a diffeomorphism from $\widetilde{\mathcal{S}} \to \Omega_\rho$, which completes the proof. $\square$

Next we have the proof of Theorem 5.8, which has a similar structure. Where arguments go through with no change, we will be brief.

Proof of Theorem 5.8. The manifolds $\widetilde{\mathcal{D}}$ and $\mathbb{CH}^{n+1}$ are both smooth manifolds of dimension $2n+2$. The injectivity of Dev^D follows from an identical argument as in the proof of Theorem 5.7. For a $[v] \in \mathbb{CH}^{n+1}$, the associated function f_v must achieve a minimum by Theorem 4.7, so there is some $x \in \widetilde{X}$ such that $f_v(x) = 0$, hence $[v]$ is in the fiber over x and in the image of Dev^D. Thus, Dev^D is surjective.

As before, the action of Γ on $\mathbb{CH}^{n+1}$ is properly discontinuous. We now identify $\mathcal{D}$ with a disk bundle. Once again we will work over $\widetilde{X}$ with $\widetilde{\mathcal{D}}$ and then pass via equivariance to $\mathcal{D}$. The fiber over a point x is given by $[w, z] \in \mathcal{V}_1(x) \oplus \mathcal{V}_2(x)$ satisfying $\|w\|_{h(x)} - \|z\|_{h(x)} < 0$ with $z \neq 0$. We can normalize z^{-1} so that we have $[w, 1]$ where $\|w\|_{h(x)} < 1$. Then the disk $B^n = \{w \in \mathbb{C}^n \mid \|w\| < 1\}$ parametrizes each fiber $\mathbb{P}D_x$ for each $x \in \widetilde{X}$. This gives the desired diffeomorphism with a disk bundle.

All that remains is to show that Dev^D is a diffeomorphism. Once again, we compute the developing map in local coordinates:

$$\begin{aligned} \widetilde{X} \times S^{2n-1} \times (0,1) &\xrightarrow{\mathrm{Dev}^S} \partial\mathbb{CH}^{n+1} \\ (z, w, t) &\longmapsto [t \cdot A_w c_1(z) + c_2(z)] \end{aligned} \tag{5.4}$$

Once again, we compute the derivative locally, taking note that $\mathcal{V}_1(z_0)^{\perp}$ has higher dimension in this case as there is an extra tangent direction to the set of negative vectors rather than the set of isotropic vectors and denoting $(0,1)$ by I:

$$T_{z_0}\widetilde{X} \times T_{w_0}S^{2n-1} \times T_{t_0}I \xrightarrow{D_{(z_0,w_0)}\operatorname{Dev}^S} \hom([v_0], \mathcal{V}_1'(z_0) \oplus \mathcal{V}_3(z_0))$$

Now it is again clear that taking the derivatives of w and t only affect $\mathcal{V}_1'(z_0)$ since they are fiber directions, and each induces a nondegenerate variation. So as before, all that remains is to check $D(T_{z_0}\widetilde{X} \to \hom([v_0], \mathcal{V}_3(z_0)))$, but this calculation is exactly the same as before and we see that the derivative map is an isomorphism, which completes the proof that Dev^D is a diffeomorphism. □

5.5 The Anosov Condition for SU(n+1,1)-Higgs Bundles

Having constructed the geometric structures, our next task is to establish the Anosov condition for this class of Higgs bundles. The first step is to establish a technical lemma which will be used in the proof of the Anosov condition for both $\mathsf{SU}(n+1,1)$-Higgs bundles, and $\mathsf{SO}_0(2,n+2)$-Higgs bundles.

Lemma 5.9. *Let E be a G-Higgs bundle over $\widetilde{X}$ for some real reductive Lie group G satisfying the following:*

1. *G induces a splitting a E into subbundles $E = P \oplus N$*
2. *For a point $x_0 \in \widetilde{X}$ and a maximal compact subgroup K that corresponds to the splitting $P(x_0) \oplus N(x_0)$ at x_0. We also require that K acts transitively on $P(x_0)$ and $N(x_0)$ and the Harmonic metric is K invariant.*

3. *There is a root $\alpha \in \mathfrak{a}^+$ such that the weight spaces for $W_\alpha = \operatorname{span} f + g$ and $W_{-\alpha} = \operatorname{span} \pm f \mp g$ for orthogonal unit vectors f and g with f in either $P(x_0)$ or $N(x_0)$, and g in whichever one that f is not in. Without loss of generality, assume that $f \in P(x_0)$. If, in fact, $f \in N(x_0)$, then all the following results hold with 'positive' and '$P(x_0)$' replaced by 'negative' and '$N(x_0)$'.*

4. *The opposition involution, induced by the longest element of the Weyl group of $\mathfrak{g}$, is trivial.*

If these conditions are satisfied, then for each $\gamma \in \Gamma$, there exists some $f' \in P(x_0)$ and uniform constants $C_1, C_2 > 0$, independent of γ, such that:

$$\|f'\|^2_{h(\gamma x_0)} \leq C_1 e^{2\alpha(\gamma)} + C_2$$

where $\alpha(\gamma)$ is shortened notation for $\alpha(\mu(\rho(\gamma)))$, where $\mu : \mathsf{G} \to \mathfrak{a}^+$ is the Cartan projection.

Proof. First, note that $2f = f + g \pm (\pm f \mp g)$. We will use exp to denote the Lie algebra exponential, and $e^{\cdot}$ for scalar exponential. From condition 3, for any $A \in \mathfrak{a}^+$, we can write:

$$2\exp(A)f = e^{\alpha(A)}(f + g) \pm e^{-\alpha(A)}(\pm f \mp g)$$

Since $\alpha \in \mathfrak{a}^+$, $-\alpha(\rho) \leq 0$. We can take the norm squared of this expression with respect to the Harmonic metric:

$$4\|\exp(A)f\|^2_{h(x_0)} = e^{2\alpha(\gamma)} + e^{-2\alpha(\gamma)} \tag{5.5}$$

using the orthogonality of f and g.

Now for any $\rho(\gamma) \in \Gamma$, we have a KAK decomposition as:

$$\rho(\gamma) = k_1 \cdot \exp(\mu(\rho(\gamma))) \cdot k_2$$

for $k_1, k_2 \in \mathsf{K}$. Since the opposition involution is trivial on G, we know

$$\rho(\gamma)^{-1} = k_2^{-1} k_0^{-1} \cdot \exp(\alpha(\gamma)) \cdot k_0 k_1^{-1}$$

for a $k_0 \in \mathsf{K}$ that induces the action of the longest element of the Weyl group. Set $k_0 k_1^{-1} = k'$. Then for any $w \in E$ and using both the definition of the harmonic metric and it's invariance under K:

$$\|w\|_{h(\gamma x_0)} = \|\rho(\gamma)^{-1} w\|_{h(x_0)} = \|\exp(\alpha(\gamma)) \cdot k' w\|$$

Now let $f' \in P(x_0)$ be such that $k' f' = f$. We have:

$$\|f'\|_{h(\gamma x_0)} = \|\exp(\alpha(\gamma)) \cdot k' f'\|_{h(x_0)} = \|\exp(\alpha(\gamma)) \cdot f\|_{h(x_0)}$$

Squaring the equation and applying Equation 5.5, we see:

$$\|f'\|^2_{h(\gamma x_0)} \leq \frac{1}{4} e^{2\alpha(\gamma)} + C$$

for any $C > 1$. Thus, we have the desired bounds that are independent of γ. $\square$

Armed with this lemma, we are able to prove another key theorem.

Theorem 5.10. *For a polystable* $\mathsf{SU}(n+1,1)$*-Higgs bundle that decomposes as*

$$\mathcal{V}_1 \xrightarrow{\beta_1} \mathcal{V}_2 \xrightarrow{1} \mathcal{V}_3$$

the associated representation $\rho : \pi_1(X) \to \mathsf{SU}(n+1,1)$ *is Anosov. Moreover,* Ω_ρ*, from Theorem 5.7, is a domain of discontinuity and the complement of the open set* Ω_ρ *in* $\partial\mathbb{CH}^{n+1}$ *coincides with the limit set* Λ_ρ*.*

Proof. Recall from Definition 3.14, showing that ρ is Anosov means that we have to establish:

$$\alpha(\mu(\rho(\gamma))) \geq \epsilon \cdot d(x_0, \gamma x_0) - C \tag{5.6}$$

for any $\gamma \in \pi_1(X)$ and where $\alpha \in \mathfrak{a}^*$ is the root and μ is the Cartan projection, $\mu : \mathsf{SU}(n+1,1) \to \mathfrak{a}^+$ and d is the hyperbolic metric on $\widetilde{X}$.

To begin, we fix $\mathsf{K} = \mathsf{S}(\mathsf{U}(n+1) \times \mathsf{U}(1))$ a maximal compact corresponding the Hermitian metric h at the base point $x_0 \in \widetilde{X}$. Then we get associated decomposition into a positive P and negative N, $P \oplus N$, orthogonal with respect to both h and h_I, where N is dimension $n+1$ and is positive with respect to h_I and N is 1 dimensional and negative with respect to h_I.

We choose compatible $\mathfrak{a}$, positive Weyl chamber $\mathfrak{a}^+$, and weight spaces with positive root α as described in Section 5.2. Notice that e_1 and e_{n+2} satisfy the conditions of Lemma 5.9, taking e_1 to be f from the lemma. Note that since e_1 is a negative root, we will use the negative version. Thus, for any $\gamma \in \Gamma$, we know that there is a negative vector w' such that:

$$\|w'\|^2_{h(\gamma x_0)} \leq \frac{1}{4}e^{2\alpha(\gamma)} + C \tag{5.7}$$

for some universal constant $C > 0$.

Take the function $f_{w'}$ as described above. Since w' is a negative vector and we proved that $f_{w'}$ satisfies the condition of Definition 4.5, we know that it satisfies an exponential growth condition of Proposition 4.8. We know that $f_{w'}(\gamma x_0) \leq \|w'\|_{\gamma x_0}$, since h is a positive definite metric. So we can combine these results into:

$$\frac{1}{C_1}e^{\epsilon \cdot d(x_0, \gamma x_0)} - C_2 \leq f_{w'}(\gamma x_0) \leq \|w'\|^2_{h(\gamma x_0)} \tag{5.8}$$

Now combining Equations 5.7 and 5.8, we get:

$$\frac{1}{C_1}e^{\epsilon \cdot d(x_0, \gamma x_0)} - C_2 \leq \frac{1}{4}e^{2\alpha(\mu(\rho(\gamma)))} + C$$

which we can rewrite into precisely the desired inequality:

$$\alpha(\mu(\rho(\gamma))) \geq \frac{1}{C_3} \cdot d(x_0, \gamma x_0) - C_4$$

Since this holds for each $\gamma \in \pi_1(X)$, we have shown the Anosov condition.

We already saw in Theorem 5.7 that the action of ρ on Ω_ρ was properly discontinuous. Now suppose that $[v] \in \partial\mathbb{CH}^{n+1} \setminus \Omega_\rho$ is isotropic. From the proof of Theorem 5.7, f_v cannot achieve a minimum on $\widetilde{X}$, else $[v]$ would be in the image of Dev^S and so in Ω_ρ. Suppose for the sake of contradiction that $[v]$ is also not contained in the limit set of ρ. Then f_v must grow in every direction of the geodesic flow on $\widetilde{X}$, which would imply that f_v is proper. However, this is a contradiction since a proper function achieves a minimum. On the other hand, if $[v]$ were in the limit set of ρ, then the geodesic flow has a direction on which f_v is bounded. This means that f_v can't be proper, and so $[v]$ can't be in Ω_ρ, else it would contradict the isotropic exponential growth condition of Theorem 4.8. Thus, the limit set of ρ is identified with Λ_ρ, the complement of Ω_ρ. $\square$

There is an immediate corollary of this theorem.

Corollary 5.11. *With the same set up as in 5.10, the representation ρ is convex cocompact.*

Proof. We saw that the Cartan subalgebra of $\mathfrak{su}(n+1,1)$ was 1 dimensional, so $\mathsf{SU}(n+1,1)$ is rank 1. Theorem 5.15 in [GW12] says that in rank 1, being Anosov is equivalent to being convex cocompact. $\square$

For $n = 1$, it is possible to parametrize a family of such Higgs bundles in terms of a line bundle L where $0 \leq \deg L \leq g-1$ and a section $\beta \in H^0(X, \hom(L^2K, L^{-1}) \otimes K) = H^0(X, L^{-3} \otimes K)$, where we recall that K is the canonical bundle. We take then take an $\mathsf{SU}(2,1)$-Higgs bundle of form:

$$L^2K \xrightarrow{\beta} L^{-1} \xrightarrow{1} L^{-1}K^{-1} \tag{5.9}$$

where β and $1 : L^{-1} \to L^{-1}K^{-1} \otimes K = L^{-1}$, are the data of the Higgs field and L^{-1} is the negative line bundle with respect to the signature $(2,1)$ Hermitian form. We see that such a bundle is stable, as the only fixed subbundle of the Higgs fields is $L^{-1}K^{-1}$ which has negative degree. As a stable Higgs bundle, it satisfies the conditions of Theorem 5.10. We summarize this in the following corollary.

Theorem 5.12. *For* $\mathsf{SU}(2,1)$ *and a choice of Riemann surface structure* X *on a surface of genus* $g \geq 2$*, there is a parametrization of a class of convex cocompact representations from Theorem 5.11 given by a line bundle* L *and a section* $\beta \in H^0(X, \hom(L^2K, L^{-1}) \otimes K)$*.*

We have an immediate consequence of this parametrization.

Corollary 5.13. *For each possible value of the Toledo invariant on the character variety of* $\mathsf{SU}(2,1)$*, there is a convex cocompact representation with that value. In particular, there is a convex cocompact representation in every component of the character variety.*

Proof. Recall that for a $\mathsf{SU}(p,q)$-Higgs bundle $(\mathcal{U} \oplus \mathcal{V}, \Phi)$, the Toledo invariant is given in terms of the degrees of $\mathcal{U}$ and $\mathcal{V}$, along with p and q:

$$\tau = 2\frac{\deg \mathcal{U} \cdot q - \deg \mathcal{V} \cdot p}{p+q}$$

For any L with $0 \leq \deg L \leq g-1$, we see that for an $\mathsf{SU}(2,1)$-Higgs bundles parametrized in the form of Equation 5.9, we get the Toledo invariant:

$$\tau = 2\frac{2\deg L - (-\deg L)}{3} = 2\deg L$$

since $\deg \mathcal{U} = \deg L$ and $\deg \mathcal{V} = -\deg L$.

Thus based on the choice of L, τ is any even integer between 0 and $2g-2$, which are all possible values of the Toledo invariant. It was shown in [Xia03] that the Toledo invariant characterizes components of the $\mathsf{SU}(2,1)$ character variety. Since every Toledo invariant is realized for Higgs bundles of this form and these correspond to convex cocompact representations, there is a convex cocompact representation in every component of the character variety. □

which recovers a result of Goldman, Kapovich and Leeb [GKL01] using very different techniques.

Chapter 6

SO(2,n+2) Higgs Bundles

In this section, we will prove the main results stemming from the specific class of $\mathsf{SO}_0(2,n+2)$-Higgs bundles described in the introduction.

6.1 SO(2,n+2) Lie Theory Basics

We start by determining a matrix H which will induce a signature $(2,n+2)$ inner product on $\mathbb{R}^{n+4}$, giving it the structure of $\mathbb{R}^{2,n+2}$. Let H be the matrix:

$$H = \begin{bmatrix} 1 & & & & \\ & 1 & & \text{\Large 0} & \\ & & -1 & & \\ & \text{\Large 0} & & \ddots & \\ & & & & -1 \end{bmatrix}$$

Now, for $u, v \in \mathbb{R}^{n+4}$, there is an inner product given by $\langle u, v \rangle = u^T H v$. Cleary, the signaute is $(2, n+2)$ as desired. The Lie group $\mathsf{SO}(2, n+2)$ is given by:

$$\mathsf{SO}(2, n+2) = \{A \in \mathsf{SL}(n+4, \mathbb{R}) \mid A^T H A = H\}$$

and we let $\mathsf{SO}_0(2, n+2)$ be the component containing the identity. An element X of the Lie algebra is an element of $\mathfrak{sl}(n+4, \mathbb{R})$ satisfying $A^T H + HA = 0$. We can decompose such an X into block form as:

$$X = \begin{bmatrix} a & b \\ b^T & d \end{bmatrix}$$

where a and d are skew symmetric matrices of dimension 2×2 and $(n+2) \times (n+2)$ respectively. We also require the condition that $\operatorname{Tr} a + \operatorname{Tr} d = 0$ in order to lie inside $\mathfrak{sl}(n+2, \mathbb{C})$. We define the Cartan involution θ to by $\theta : X \mapsto -X^T$. Then:

$$\theta(X) = \begin{bmatrix} -a^T & -b \\ -b^T & -d^T \end{bmatrix} = \begin{bmatrix} a & -b \\ -b^T & d \end{bmatrix}$$

where we made use of the skew symmetry of a and d. We find that the fixed points are matrices with $b = 0$, whereas matrices with $a = 0$ and $d = 0$ are multiplied by -1. Hence the ± 1-eigenspaces of the Cartan decomposition are given by:

$$\mathfrak{l} = \left\{ \begin{bmatrix} a & b \\ b^T & d \end{bmatrix} \in \mathfrak{so}(2, n+2) \;\middle|\; b = 0 \right\} = \left\{ \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \in \mathfrak{so}(2, n+2) \right\}$$

$$\mathfrak{p} = \left\{ \begin{bmatrix} a & b \\ b^T & d \end{bmatrix} \in \mathfrak{so}(2, n+2) \;\middle|\; a = 0, d = 0 \right\} = \left\{ \begin{bmatrix} 0 & b \\ b^T & 0 \end{bmatrix} \in \mathfrak{so}(2, n+2) \right\}$$

The Cartan subalgebra $\mathfrak{a}$ is given by matrices A in $\mathfrak{p}$ where:

$$\begin{bmatrix} 0 & a_2 & 0 \cdots\cdots\cdots 0 \\ a_1 & 0 & 0 \cdots\cdots\cdots 0 \end{bmatrix}$$

Let f_i be the linear functional in $\mathfrak{a}^*$ where $f_i(A) = a_i$ for $i = 1, 2$. Then the restricted root system is $\Phi = \{\pm f_1 \pm f_2, \pm f_1, \pm f_2\}$. We choose a basis of simple roots $\Delta = \{\alpha_1 = f_1 - f_2, \alpha_2 = f_2\}$. Then the Weyl chamber $\mathfrak{a}^+$ is matrices $A \in \mathfrak{a}$ where $a_1 \geq a_2 \geq 0$.

Ultimately, we wish to compute the weight spaces for the standard representation $\mathfrak{a} \to \mathfrak{gl}(n+4, \mathbb{R})$. In order to do so, we will compute a change of basis so that matrices in $\mathfrak{a}$ have the form:

$$A = \begin{bmatrix} a_1 & & & & \\ & a_2 & & & \\ & & 0 & & \\ & & & -a_2 & \\ & & & & -a_1 \end{bmatrix}$$

Under this change of basis, H becomes:

$$H = \begin{bmatrix} & 0 & & & -1 \\ & & & -1 & \\ & & -\mathrm{Id}_n & & \\ & -1 & & & \\ -1 & & & 0 & \end{bmatrix}$$

Where we use H to denote $V^{-1}HV$, H under the change of basis. There is an orthonormal basis, $e_1, \ldots, e_{n+4}$, with positive vectors e_1, e_2 and the rest are negative. In this basis, the weight space for α_2 is given by $\operatorname{span} e_{n+3} - e_2$, and the weight space for $-\alpha_2$ is $\operatorname{span} e_{n+3} + e_2$. This will be vital in applying Lemma 5.9 in the proof of Theorem 6.6.

6.2 SO(2,n+2) Higgs Bundles

We recall from Definition 3.8 that an $\mathsf{SO}_0(2, n+2)$-Higgs bundle over a Riemann surface X with genus $g \geq 2$ is a tuple (V, W, Q_V, Q_W, η), where V is a rank 2 holomorphic vector bundle and W is rank $n+2$. We shall restrict our attention to a class of polystable $\mathsf{SO}_0(2, n+2)$-Higgs bundles that are parametrized by: $(\mathcal{W}_3, \mathcal{W}_4, \alpha)$ where:

- $\mathcal{W}_3$ is a degree 0, rank n holomorphic vector bundle over X
- $\mathcal{W}_4$ is a line bundle over X
- α is a section in $H^0(X, \hom(\mathcal{W}_3, \mathcal{W}_4) \otimes K)$

We then define the line bundle $\mathcal{W}_2 = \overline{\mathcal{W}_4}$, and line bundles $\mathcal{W}_5 = \mathcal{W}_4 K^{-1}$ with $\mathcal{W}_1 = \overline{\mathcal{W}_5}$ such that

$$\Lambda^{n+4}(\mathcal{W}_1 \oplus \mathcal{W}_2 \oplus \mathcal{W}_3 \oplus \mathcal{W}_4 \oplus \mathcal{W}_5) = \mathcal{O}$$

This gives us a Higgs bundle:

$$\mathcal{W}_1 \xrightarrow{1} \mathcal{W}_2 \xrightarrow{\alpha^T} \mathcal{W}_3 \xrightarrow{\alpha} \mathcal{W}_4 \xrightarrow{1} \mathcal{W}_5$$

where α^T makes sense because $\alpha \in H^0(X, \mathrm{hom}(\mathcal{W}_3, \mathcal{W}_4) \otimes K)$, and the adjoint with respect to the harmonic metric is the conjugate transpose: $\alpha^* \in H^0(X, \mathrm{hom}(\mathcal{W}_4, \mathcal{W}_3) \otimes K)$, so $\alpha^T \in H^0(X, \mathrm{hom}(\overline{\mathcal{W}_4}, \mathcal{W}_3) \otimes K)$ and using the definition of $\mathcal{W}_2$, we get $\alpha^T \in H^0(X, \mathrm{hom}(\mathcal{W}_2, \mathcal{W}_3) \otimes K)$. Similarly for 1, which is taken to be an isomorphism.

We will require that the positive definite component of the Higgs bundle is $\mathcal{W}_2 \oplus \mathcal{W}_4$ and the negative definite componenent is $\mathcal{W}_1 \oplus \mathcal{W}_3 \oplus \mathcal{W}_5$ and that this decomposition is orthogonal with respect to the harmonic metric.

As in Chapter 5, we note that the flat connection decomposes into:

$$\nabla = \nabla^{Ch} + \varphi + \varphi^*$$

For a flat section v, we similarly see that

$$0 = (\nabla v)_5 = \nabla^{Ch} v_5 + \varphi(v_4) = \nabla^{Ch} v_5 + \alpha(v_4)$$

We will use this fact to construct another function which satisfies the conditions of Chapter 4. Let $E_{\mathbb{R}}$ be the real subbundle of E, that is, all the fixed points of conjugation on the fibers. Fix a basepoint $x_0 \in X$ and let $E_{\mathbb{R}}(x_0)$ be the real vector subspace of the fiber over x_0. Then pullback the bundle over the universal cover of X, and use the flat connection to trivialize the pullback into $\widetilde{X} \times E_{\mathbb{R}}(x_0)$.

For a nonnegative vector $v \in E_{\mathbb{R}}(x_0) \setminus 0$ with respect to the indefinite hermitian metric, it extends using the trivial connection on the trivialization to a flat section. In terms of the above decomposition, it splits into $v = v_1 + v_2 + v_3 + v_4 + v_5$, for $v_i \in \mathcal{W}_i$. In particular, due to the condtions $\mathcal{W}_1 = \overline{\mathcal{W}_5}$ and $\mathcal{W}_2 = \overline{\mathcal{W}_5}$, we compute:

$$\begin{aligned}
\|v\|_{h_I(x_0)} &= -\|v_1\|_{h(x_0)} + \|v_2\|_{h(x_0)} - \|v_3\|_{h(x_0)} + \|v_4\|_{h(x_0)} - \|v_5\|_{h(x_0)} \\
&= 2\|v_4\|_{h(x_0)} - \big(\|v_3\|_{h(x_0)} + 2\|v_5\|_{h(x_0)}\big) \\
&> 0
\end{aligned}$$

So $\|v_4\|_{h(x_0)} \geq 1$ and $\|v_4\|_{h(x_0)} \geq \|v_5\|_{h(x_0)}$.

Definition 6.1. Let $v \in E_{\mathbb{R}}(x_0)$ be a non-zero positive or isotropic vector. We define $f_v : \widetilde{X} \to [0, \infty)$ by $f_v(x) = \|v_5(x)\|_{h(x)}$. So $f_v(x)$ is the norm of the v_5 component with respect to the metric x.

Once, again, as $1 : \mathcal{W}_4 \to \mathcal{W}_5 \otimes K$ is an isomorphism and thus non-vanishing, we can define the *directing vector field* V_v on X such that

$$1(V_v)v_4 = -v_5$$

Lemma 6.2. *For a nonzero nonnegative vector $v \in E_{\mathbb{R}}(x_0)$, the function f_v satisfies conditions (C1) and (C2) of defintion 4.5. If v is positive, then it also satisfies (C3).*

The proof is exactly the same as in Lemma 5.3 so we will not recreate it.

6.3 Geometric Structures

As in the previous chapter, we shall construct two bundles over X and equip them with geometric structures. The first $(\mathsf{G}, X)-$geometry is an *Einstein structure* with $\mathsf{G} = \mathsf{SO_0}(2, n+2)$ and $X = \mathrm{Ein}^{1,n+1}$ where $\mathrm{Ein}^{1,n+1}$ is the set of isotropic lines in $\mathbb{R}^{2,n+2}$.

Consider the following decomposition of a complex vector bundle, $E = \mathcal{U} \oplus \mathcal{W}$ where $\mathcal{U} = \mathcal{W}_2 \oplus \mathcal{W}_4$ is the rank 2 positive component and $\mathcal{V} = \mathcal{W}_1 \oplus \mathcal{W}_3 \oplus \mathcal{W}_5$ is the rank $n+2$ negative component. In this, $\mathcal{W}_1$ and $\mathcal{W}_5$ are both line bundles, and $\mathcal{W}_3$ is a rank n bundle.

$$E = \mathcal{W}_1 \oplus \mathcal{W}_2 \oplus \mathcal{W}_3 \oplus \mathcal{W}_4 \oplus \mathcal{W}_5$$

We set $E_{\mathbb{R}}$ to be the real vector subbundle of E. Over a point $x \in \widetilde{X}$, we define

$$B_x = \{v \in E(x_0)_{\mathbb{R}} \mid \|v\|_{h_I(x_0)} = 0 \text{ and } v \perp \mathcal{W}_5(x)\}$$

So B_x are the real isotropic vectors in the fiber over x that live entirely in $(\mathcal{W}_2(x) \oplus \mathcal{W}_3 \oplus \mathcal{W}_4)_{\mathbb{R}}$, that is, they lie in a copy of $\mathbb{R}^{2,n}$. So for each $\mathbb{P}B_x \cong \mathrm{Ein}^{1,n-1}$, although the $\mathbb{P}B_x \subset \mathrm{Ein}^{1,n+1}$. Define the bundle $\widetilde{\mathcal{B}}$ to be the bundle over $\widetilde{X}$ whose fiber at each x is $\mathbb{P}B_x$. Then $\widetilde{\mathcal{B}} \subset \widetilde{X} \times \mathrm{Ein}^{1,n+1}$ is a $\mathrm{Ein}^{1,n-1}$-bundle.

A pseudosphere $\mathbb{S}^{1,n+1}$ in $\mathbb{R}^{2,n+2}$ is the set of all vectors of unit norm. We will construct a bundle similar to $\widetilde{\mathcal{B}}$ by projectivizing pseudospheres. With the same decomposition of E as above, we define:

$$P_x = \{v \in E(x_0)_{\mathbb{R}} \mid \|v\|_{h_I(x_0)} = 1 \text{ and } v \perp \mathcal{W}_5(x)\}$$

So P_x is the pseudosphere inside $(\mathcal{W}_2(x) \oplus \mathcal{W}_3(x) \oplus \mathcal{W}_4(x))_{\mathbb{R}}$. As above, we define $\widetilde{\mathcal{P}}$ to be the bundle over $\widetilde{X}$ where each fiber is $\mathbb{P}P_x$. The total space of the $\mathbb{P}\mathbb{S}^{1,n-1}$-bundle is then $\widetilde{\mathcal{P}} \subset \widetilde{X} \times \mathbb{P}\mathbb{S}^{1,n+1}$.

Both of these bundles descend to bundles over X.

Proposition 6.3. *The action of $\pi_1(X)$ on $\widetilde{X} \times \mathbb{P}\mathbb{S}^{1,n+1}$ and $\widetilde{X} \times \mathrm{Ein}^{1,n+1}$, where the action on the second factor is given by the representation ρ, leaves $\widetilde{\mathcal{S}}$ and $\widetilde{\mathcal{D}}$, respectively, invariant. Hence there exists bundles:*

$$\mathcal{B} := \left(\widetilde{\mathcal{B}}/\pi_1(X)\right) \to \left(\widetilde{X}/\pi_1(X)\right) = X$$
$$\mathcal{P} := \left(\widetilde{\mathcal{P}}/\pi_1(X)\right) \to \left(\widetilde{X}/\pi_1(X)\right) = X$$

Proof. The proof is identical the the proof of Proposition 5.6 as $\mathsf{SO}_0(2, n+2)$ preserves a signature $(2, n+2)$ metric, so it will preserve isotropic lines and positive lines. □

Now we have the following maps given by projection onto the second factor of the total space:

$$\mathrm{Dev}^B : \widetilde{\mathcal{B}} \to \mathrm{Ein}^{1,n+1}$$

$$\mathrm{Dev}^P : \widetilde{\mathcal{P}} \to \mathbb{P}\mathbb{S}^{1,n+1}$$

Now we can give these bundles the appropriate geometric structures by showing that maps Dev^B and Dev^P are developing maps. Associated to the Higgs bundle which induced these structures is a representation ρ. We use the notation $\Gamma = \rho(\pi_1(X))$.

Theorem 6.4. *There exists a unique open set $\Omega_\rho \subset \mathrm{Ein}^{1,n+1}$ such that Dev^B is a diffeomorphism between $\widetilde{\mathcal{B}}$ and Ω_ρ. Then Dev^B is a developing map and defines a $\mathrm{Ein}^{1,n+1}$ structure on $\mathcal{B}$.*

Moreover, the action of Γ on Ω_ρ is properly discontinuous, and Ω_ρ/Γ is diffeomorphic to $\mathcal{B}$, where in turn, $\mathcal{B}$ is diffeomorphic to a S^{n-1}-bundle.

Theorem 6.5. *The map Dev^P is a diffeomorphism between $\widetilde{\mathcal{P}}$ and $\mathbb{PS}^{1,n+1}$ and so Dev^P is a developing map which gives $\mathcal{P}$ a $\mathbb{PS}^{1,n+1}$ geometric structure.*

Additionally, Γ acts properly discontinuously on $\mathbb{PS}^{1,n+1}$ and $\mathbb{PS}^{1,n+1}$ is diffeomorphic to $\mathcal{P}$.

Proof of Theorem 6.4. The smoothness of the developing map is apparent as a projection from a smooth product manifold to a smooth manifold. For a $[v] \in \mathrm{Ein}^{1,n+1}$, v is an isotropic vector. If $[v]$ is in the image of Dev^B, then there is some x such that $v \perp \mathcal{W}_5(x)$. This is equivalent to the function $f_v(x) = 0$. This can happen for at most one $x \in \widetilde{X}$ by Theorem 4.7. Hence Dev^S is injective.

We denote the image of Dev^B by Ω_ρ (here the ρ is the representation associated to the Higgs bundle which induced Dev^B). Γ acts properly discontinuously on Ω_ρ because the action of $\pi_1(X)$ on $\widetilde{X}$ is properly discontinuous and the developing map is $\pi_1(X)$ equivariant.

Our next step is identifying $\mathcal{B}$ with an S^{n-1}-bundle. For any $x \in \widetilde{X}$, choose a $w \in (W_5(x))_{\mathbb{R}}$ such that $\|w\|_h = 1$. Then

$$v = \frac{1}{\sqrt{2}} \oplus w \oplus \frac{1}{\sqrt{2}}$$

is an element of the fiber of $\widetilde{\mathcal{B}}$ over x. This gives an identification between S^{n-1} and each fiber of $\widetilde{\mathcal{B}}$, hence the desired identification with a S^{n-1}-bundle.

Our last step is to verify that Dev^B is a diffeomorphism. In a local chart of $\widetilde{X}$ with coordinate z, we take s_0 to be a local unit-norm section of $\mathcal{W}_4$, and $s_1, \dots s_n$ to be local orthonormal frame of $\mathcal{W}_3$. Then:

$$\begin{aligned} \widetilde{X} \times S^{n-1} &\xrightarrow{\mathrm{Dev}^B} \mathrm{Ein}^{1,n+1} \\ (z, w) &\longmapsto \overline{s_0(z)} + \sum_{i=1}^{n} w_i s_i(z) + s_0(z) \end{aligned} \tag{6.1}$$

which lies in $\mathcal{W}_2(z) \oplus \mathcal{W}_3(z) \oplus \mathcal{W}_4(z)$.

Working a neighborhood of a point z_0 in our local coordinate, we define:

$$N = (\mathcal{W}_1(z_0) \oplus \mathcal{W}_5(z_0))_{\mathbb{R}}, \quad P = (\mathcal{W}_2(z_0) \oplus \mathcal{W}_4(z_0))_{\mathbb{R}}, \quad O = (\mathcal{W}_3(z_0))_{\mathbb{R}}$$

So our decomposition will be in terms of $N \oplus O \oplus P$.

Consider a $[v_0] \in \mathrm{Ein}^{1,n+1}$ in the image of the developing map on the fiber over z_0. So $[v_0] = \mathrm{Dev}^B(z_0, w_0)$. Let $P' = P \cap [v_0]^{\perp}$, and $O' = O \cap [v_0]^{\perp}$. We see that $P' = 0$ since $[v_0]$ is an isotropic vector and so must have a fixed $\mathcal{W}_4$ component.

Thus $[v_0]^\perp \cong N \oplus O'$, so

$$T_{[v_0]} \operatorname{Ein}^{1,n+1} = \hom([v_0], [v_0]^\perp) = \hom([v_0], N \oplus O')$$

In local coordinates, the derivative of the developing map is given by:

$$T_{z_0}\widetilde{X} \times T_{w_0} S^{n-1} \xrightarrow{D_{(z_0,w_0)} \operatorname{Dev}^B} \hom([v_0], N \oplus 0')$$

Now we can decompose this into block form as:

$$\begin{bmatrix} D(T_{z_0}\widetilde{X} \to \hom([v_0], N) & 0 \\ D(T_{z_0}\widetilde{X} \to \hom([v_0], O') & D(T_{w_0}S^{n-1} \to \hom([v_0], O') \end{bmatrix}$$

It is clear from Equation 6.1 that perturbing w_0 must leave the image inside $O \oplus P$, so the top right entry is 0. Differentiating in the w_0 direction clearly induces a nontrivial variation in the O' direction, so the bottom right entry is non-degenerate. Finally, we will compute $D(T_{z_0}\widetilde{X} \to \hom([v_0], N)$ using the flat connection modulo O and P.

$$\nabla s_0 = \nabla^{Ch} s_0 + 1(s_0) + \alpha^*(s_0) = 1(s_0) \mod O \oplus P$$

which is nontrivial as s_0 is a non-vanishing section and 1 is an isomorphism. Computing ∇s_i mod $O \oplus P$ all give 0 for $i = 1, \ldots, n$, hence the derivative is nondegenerate and Dev^B is a diffeomorphism onto its image.

□

Proof of Theorem 6.5. As was the case in Chapter 5 with Theorems 5.7 and 5.8, these two theorems have very similar proofs. The considerations which need to be taken between this proof and the proof of Theorem 6.4 are the same as for the analagous theorems in Chapter 5, so we will skip a fourth recreation of this style of proof. ☐

6.4 The Anosov Condition for SO(2,n+2) Bundles

Theorem 6.6. *Let E be a polystable $\mathsf{SO}_0(2, n+2)$-Higgs bundle that decomposes as*

$$\mathcal{W}_1 \longrightarrow \mathcal{W}_2 \longrightarrow \mathcal{W}_3 \xrightarrow{\alpha} \mathcal{W}_4 \xrightarrow{1} \mathcal{W}_5$$

Then the associated representation $\rho : \pi_1(X) \to \mathsf{SO}_0(2, n+2)$ is α_2-Anosov and Ω_ρ from Theorem 6.4 is a domain of discontinuity.

Proof. From Section 6.1, the vectors e_2 and e_{n+3} satisfy the conditions of Lemma 5.9 for the root α_2, once we fix an appropriate maximal compact and choice of positive Weyl chamber. Then for any $\gamma \in \Gamma$, we know that there is a positive vector w' such that:

$$\|w'\|^2_{h(\gamma x_0)} \leq \frac{1}{4} e^{2\alpha_2(\gamma)} + C \tag{6.2}$$

for some universal constant $C > 0$. Here, $\alpha_2(\gamma)$ is shorthand for $\alpha_2(\mu(\rho(\gamma)))$.

Take the function $f_{w'}$ from Definition 6.1. Since w' is a positive vector and we proved that $f_{w'}$ satisfies the condition of Definition 4.5 in Lemma 6.2, we know that it satisfies an exponential growth condition of Proposition 4.8. We know that $f_{w'}(\gamma x_0) \leq \|w'\|_{\gamma x_0}$, since

h is a positive definite metric. From this point on, the proof follows the exact steps of the proof of Theorem 5.10, and we get the same results. $\square$

Apart from the results themselves, a key takeaway is how powerful functions of the type described in Definitions 5.2 and 6.1 are. Not only do they satisfy many properties, but they interact with both the data of the Lie group and representation. Standardizing these results across a wider range of Higgs bundles and scenarios seems to be a promising direction for further investigation.

References

[ADL24] Daniele Alessandrini, Colin Davalo, and Qiongling Li, *Projective structures with (quasi-)Hitchin holonomy*, J. Lond. Math. Soc. (2) **110** (2024), no. 4, Paper No. e13003, 56. MR4803942

[Ale19] Daniele Alessandrini, *Higgs bundles and geometric structures on manifolds*, SIGMA Symmetry Integrability Geom. Methods Appl. **15** (2019), Paper 039, 32. MR3948927

[Bar09] D Baraglia, *G2 geometry and integrable systems*, Ph.D. Thesis, 2009.

[BGPG03] Steven B. Bradlow, Oscar García-Prada, and Peter B. Gothen, *Surface group representations and* $\mathrm{U}(p,q)$*-Higgs bundles*, J. Differential Geom. **64** (2003), no. 1, 111–170. MR2015045

[Bis15] James Bisgard, *Mountain passes and saddle points*, SIAM Rev. **57** (2015), no. 2, 275–292. MR3345345

[CG93] Suhyoung Choi and William M. Goldman, *Convex real projective structures on closed surfaces are closed*, Proc. Amer. Math. Soc. **118** (1993), no. 2, 657–661. MR1145415

[Cor88] Kevin Corlette, *Flat g-bundles with canonical metrics*, J. Differential Geom. **28** (1988), no. 3, 361–382. MR965220

[CTT19] Brian Collier, Nicolas Tholozan, and Jérémy Toulisse, *The geometry of maximal representations of surface groups into so0(2,n)*, Duke Mathematical Journal **168** (October 2019), no. 15.

[Dav24] Colin Davalo, *Geometric structures for maximal representations and pencils*, 2024.

[Don87] S. K. Donaldson, *Twisted harmonic maps and the self-duality equations*, Proc. London Math. Soc. (3) **55** (1987), no. 1, 127–131. MR887285

[Fil22] Simion Filip, *Uniformization of some weight 3 variations of hodge structure, anosov representations, and lyapunov exponents*, 2022.

[GKL01] William M. Goldman, Michael Kapovich, and Bernhard Leeb, *Complex hyperbolic manifolds homotopy equivalent to a Riemann surface*, Comm. Anal. Geom. **9** (2001), no. 1, 61–95. MR1807952

[Gol22] William M. Goldman, *Geometric structures on manifolds*, Graduate Studies in Mathematics, vol. 227, American Mathematical Society, Providence, RI, 2022. MR4500072

[GPGMiR09] Oscar García-Prada, Peter B. Gothen, and Ignasi Mundent i Riera, *he hitchin-kobayashi correspondence, higgs pairs and surface group representation*, 2009.

[GW08] Olivier Guichard and Anna Wienhard, *Convex foliated projective structures and the Hitchin component for* $\mathrm{PSL}_4(\mathbf{R})$, Duke Math. J. **144** (2008), no. 3, 381–445. MR2444302

[GW12] ———, *Anosov representations: domains of discontinuity and applications*, Inventiones mathematicae **190** (February 2012), no. 2, 357–438.

[Hit87a] N. J. Hitchin, *The self-duality equations on a riemann surface*, Proc. London Math. Soc. (3) **55** (1987), no. 1, 59–126. MR887284

[Hit87b] Nigel Hitchin, *Stable bundles and integrable systems*, Duke Math. J. **54** (1987), no. 1, 91–114. MR885778

[Hit92] N. J. Hitchin, *Lie groups and Teichmüller space*, Topology **31** (1992), no. 3, 449–473. MR1174252

[KL05] Bruce Kleiner and Bernhard Leeb, *Rigidity of invariant convex sets in symmetric spaces*, Inventiones mathematicae **163** (October 2005), no. 3, 657–676.

[KLP16a] Michael Kapovich, Bernhard Leeb, and Joan Porti, *Some recent results on Anosov representations*, Transform. Groups **21** (2016), no. 4, 1105–1121. MR3569569

[KLP16b] ———, *Some recent results on Anosov representations*, Transform. Groups **21** (2016), no. 4, 1105–1121. MR3569569

[KLP17] ———, *Dynamics on flag manifolds: domains of proper discontinuity and cocompactness*, Geometry and Topology **22** (2017oct), no. 1, 157–234.

[Lab06] Fran¸cois Labourie, *Anosov flows, surface groups and curves in projective space*, Invent. Math. **165** (2006), no. 1, 51–114. MR2221137

[Sim88] Carlos T. Simpson, *Constructing variations of Hodge structure using yang-mills theory and applications to uniformization*, J. Amer. Math. Soc. **1** (1988), no. 4, 867–918. MR944577

[Str96] Michael Struwe, *Variational methods*, Second, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) [Results in Mathematics and Related Areas (3)], vol. 34, Springer-Verlag, Berlin, 1996. Applications to nonlinear partial differential equations and Hamiltonian systems. MR1411681

[Xia03] Eugene Z. Xia, *The moduli of flat* $\mathrm{U}(p,1)$ *structures on Riemann surfaces*, 2003, pp. 33–43. Special volume dedicated to the memory of Hanna Miriam Sandler (1960–1999). MR2003688

[Zha24] Junming Zhang, *Non-maximal anosov representations from surface groups to* $\mathrm{SO}_0(2,3)$, 2024.

[Zhu20] Feng Zhu, *Relatively dominated representations*, 2020.